Supplementary Information for

Hydrogen in residential heating and the power system: a merit-order and capital-recovery assessment for 29 European markets

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This Supplementary Information provides the full methodology, the input parameter and scenario tables, the per-country and per-arena results and the sensitivity analyses that the main paper summarises at a high level. Section S1 gives the complete model formulation; Section S2 gives the extended results and robustness analysis. Symbols follow the nomenclature listed in the main paper.

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S1 Materials and methods

S1.1 Model overview and architecture

We assess hydrogen's role in European building heat with a spatially resolved, bottom-up techno-economic model that simulates residential heat supply demand-first and couples it to a merit-order, unit-commitment representation of the power sector. The model chains its stages, and no stage is co-optimised with another. It reconstructs useful heat demand region by region, splits that demand across eight heating technologies, prices each technology, and then prices hydrogen against its rivals in three successive tests (levelised cost, winter merit-order dispatch, and peaker cost recovery). Hydrogen's heating share is in this sense bounded. The bound comes from a merit-order *screening*, a sequence of pairwise short-run-marginal-cost threshold tests, not from a co-optimised system equilibrium; storage sizing, demand-side flexibility and capacity expansion are exogenous inputs to the screening, fixed before it runs. We frame the contribution accordingly, and return to the bounds this places on the firm-capacity verdict in Section S1.6. A parallel least-cost linear program (Section S1.7) provides a cost-optimal cross-check. Our contribution is in the first place methodological, and it also identifies a bounded exception to a comparison the review literature treats as settled.

The thermodynamic ranking is not in question. A heat pump delivers several units of heat per unit of electricity and a boiler delivers less than one unit of heat per unit of fuel, and nothing here disturbs that. The cost ranking is another matter. When equipment capital is sized to peak heat demand and annualised over the heat that capacity delivers, when each carrier is charged the last-mile network its own delivery requires, and when hydrogen is charged the seasonal store its winter-weighted load requires (Section S1.3), the 2050 base-load levelised cost of a hydrogen boiler falls below that of the best available heat pump in 7 of the 29 markets. We report that outcome, and we report the two reasons for it, which are that cheap cavern storage is confined to particular geologies and that the comparison sets taxed retail electricity against hydrogen priced delivered and untaxed. The consensus we qualify [1, 2] is a consensus about technologies, and the seven-market exception reported here is about storage geology and about levies, and we read it that way throughout.

What is additionally new is the resolution at which the bound is drawn (hydrogen's heating share is bounded country by country, from a single demand-to-dispatch chain) and the reframing of the hydrogen-for-heat question from an energy-share to a firm-capacity problem, screened consistently across three arenas (the building winter peak, district heat and the power sector) in which capital recovery for a low-utilisation peaker binds before delivered energy cost does.

We resolve demand to the third level of the Nomenclature of Territorial Units for Statistics (NUTS3), covering 1,369 regions across the EU-27, the United Kingdom and Switzerland [3], and aggregate to the country before economic tests are applied. The model runs on the milestone years 2025, 2030, 2040 and 2050, with 2025 as the base year and 2050 as the target year; all trajectories originate in 2025 so that demand, technology-share and cost ramps share a common anchor.

The architecture is demand-first, and Fig. 1 of the main text sets out both the pipeline and, in its header band, the three input layers that feed it: the building-stock and demand definition, the techno-economic parameters, and the power-side and storage inputs. Two open databases carry the bottom-up check, and neither is named in that figure: EUBUCCO, a database of European building footprints, and TABULA, a set of national building-archetype heat intensities. We establish a NUTS3 useful-heat surface first, attach building-type shares to it next, and apply technology shares last. Demand is built along two independent strands that we reconcile against each other: a top-down baseline that takes NUTS3 heat demand from the Hotmaps regional database [3] and attaches census-derived building-type shares [4, 5], and a bottom-up build that reconstructs residential heat from open building footprints and a harmonised archetype typology. The two agree to $-0.8\,\%$ across the EU-27, the United Kingdom and Switzerland and to within $\pm 25\,\%$ in 28 of 29 countries (Section S1.2).

We report all headline results on the bottom-up demand basis and retain the Hotmaps top-down estimate as an independent reconciliation benchmark. The single reconciled demand surface then propagates through the shared technology, cost and policy layers and through the scenario Monte Carlo, so the merit-order and missing-money tests that carry the hydrogen findings sit at the end of one common chain. NUTS3 boundaries and urban/rural, coastal and mountain typologies come from the Eurostat GISCO service [6] at 1:1 million scale, coordinate reference system EPSG:4326. Per-country reconstruction detail and the building-stock derivation are given in the public repository (Data availability).

S1.2 Demand reconstruction and validation

Base-year useful heat demand for space heating and hot water is the NUTS3 region/building-type surface of Section S1.1. The raw Hotmaps regional total is 3,863 TWh/yr (reference year ≈2015), which resolves to 3,858 TWh on the harmonised all-class benchmark used for validation; we quote the raw figure as the disaggregation input and the all-class figure as the 2015 comparison point. We project demand forward from $t_0 = 2025$ with a per-country, multi-driver ratio identity, and we separately attribute the historical 2015 → 2025 change additively across the same drivers with a Logarithmic Mean Divisia Index (LMDI) decomposition [7]. The two roles are distinct; the forward projection is the multiplicative driver-ratio form of Eq. (S1), whereas the LMDI is the attribution method applied to the base-period change. We adopt the LMDI for that attribution because it is exact, factor-neutral and path-independent, leaves no residual term and weights each driver uniquely. National useful heat demand evolves as the 2025 baseline scaled by population growth and household-size decline (occupancy), both of which add demand, set against envelope-renovation intensity, which removes it:

$$D_c(t) = D_c^0 \, \frac{\mathrm{Pop}_{c,t}}{\mathrm{Pop}_{c,25}} \, \frac{(\mathrm{Dw/Pop})_{c,t}}{(\mathrm{Dw/Pop})_{c,25}} \, \bigl(1 - r_{c,s}\bigr)^{t-t_0}, \qquad t \geq t_0,\ t_0 = 2025, \tag{S1}$$

where D_c^0 is national base-year demand, $\mathrm{Pop}_{c,t}$ is projected population (Eurostat EUROPOP2023 for the EU-27, the ONS 2022-based projection for the UK, the BFS Referenzszenario A-00-2025 for Switzerland), $(\mathrm{Dw/Pop})_{c,t}$ is dwellings per capita, and $r_{c,s}$ is the per-country, per-scenario envelope-renovation rate. Average dwelling size is held flat, since national series drift by less than 2 % per decade. The projection makes the demand reduction an emergent quantity, the net of demographic growth against envelope renovation, with no 2050 target imposed on it. The parameter that varies by scenario is $r_{c,s}$, sampled $r_{c,s} \sim \mathcal{N}(\mu_{c,s}, \sigma_{c,s})$ (censored to $[0, 5\,\%]$/yr) in the Monte Carlo, with central values anchored on each country's current envelope-renovation rate (EU stock-weighted ≈0.55 to 0.7 %/yr; 8, 9).

We validate the build against the 2015 Hotmaps benchmark through a vintage-matched backcast, so that the two independent estimates can be compared once the roughly decade-long vintage gap is removed. The backcast is deliberately not the inverse of the forward identity of Eq. (S1). It scales the 2025 snapshot by dwelling-stock growth alone, at a per-country rate running from 0.15 to 2.5 per cent a year and compounding to an aggregate factor of 0.924 over the decade, and it carries no population, occupancy or envelope term. The reason is that the quantity being matched is a stock vintage, and the stock series is the one input the two estimates share. The vintage-matched deviation against the Hotmaps benchmark H_c^{2015} is

$$\Delta_c^{\mathrm{VM}} = \frac{D_c(2015) - H_c^{2015}}{H_c^{2015}}. \tag{S2}$$

Across the EU-27+UK+CH aggregate the four drivers of Eq. (S2) contribute approximately +1.8 % (population), +4.6 % (occupancy), 0 % (dwelling size) and −6.3 % (envelope retrofit) to the 2015 →

2025 change,[1] so the two stock-side drivers (+6.4 %) are matched almost exactly by the envelope term and the net change is only +0.1 %.

The raw aggregate bottom-up-versus-Hotmaps deviation is −0.8 %, with 28 of 29 countries within ±25 % (Malta the sole exception). Once the roughly decade-long vintage gap is removed, the matched-vintage aggregate deviation is −8.3 %; the bottom-up backcast to 2015 totals 3,537 TWh against the Hotmaps 3,858 TWh. Both figures are on the harmonised all-class benchmark of Section S1.1. The validation table carries the all-class column, whereas the reconciliation backcast table carries the raw Hotmaps one, on which the same comparison reads −8.5 % and the per-country deviations run wider, so the two should not be read against each other without matching the benchmark basis first.

Counts and an aggregate both hide the size of a country miss, the first by discarding it and the second because the country deviations carry opposite signs and cancel. On the raw basis the mean absolute deviation across the 29 countries is 11.6 %, the demand-weighted mean 9.8 %, the root-mean-square deviation 14.4 percentage points and the median absolute deviation 8.5 %, the largest single miss being 35.6 % at Malta. On the vintage-matched basis the same four read 12.4, 10.7, 15.2 and 9.6 %. The −0.8 % aggregate is thus below a tenth of the median country deviation, which is why we read the study-area aggregate as the supported quantity and country-level verdicts as indicative.

The LMDI net change and the backcast's stock-growth factor are on different constructions. The +0.1 % is the net of four drivers that very nearly cancel, and the −7.6 % the backcast applies is stock growth alone, uncancelled because the backcast carries no envelope term to offset it. The vintage-matched deviation therefore sits below the raw one by construction, and it is the more conservative of the two statistics. We report both, and we read the aggregate agreement off the raw comparison, on which the two independent estimates are built to the same vintage.

On a vintage-matched basis the per-country deviations range from −32 % (Luxembourg) and −28 % (the United Kingdom) at the low end to about +20 % at the high end (Slovenia +20 %, Latvia +18 %, Bulgaria +17 %, Romania +13 %); the larger $\gtrsim +30\,\%$ figure arises only on the raw basis and for Malta alone (raw +35.6 %), before vintage matching; no country reaches +40 % on either basis.

An independent Eurostat-derived 2015 final-energy reference (climate-corrected, ∼ 2,075 TWh) is only ∼ 54 % of the Hotmaps calculated useful demand (3,863 TWh). The prebound and performance-gap effect runs in the direction of a shortfall, and metered stock studies find a median gap of about −11 per cent, reaching −40 per cent in the worst-performing label band [10]. That is smaller than the −46 per cent here, so the effect is a contributing factor and not a full account of the gap; the remainder we do not decompose. Converting the Eurostat final-energy figure to useful energy through the installed boiler stock would lower it further, so the useful-versus-final-energy conversion can only widen the discrepancy; the residual, which also absorbs stock-coverage differences, is not fully decomposed here. We treat this factor-of-roughly-two gap as a genuine, unresolved uncertainty in the base-year demand level (not a settled correction) and note that it propagates to every downstream cost and emissions level, though the technology ranking that carries our conclusions is set by relative per-MWh costs, which the absolute demand anchor does not enter into.

[1] The aggregate contributions are base-year-demand-weighted means of the per-country design decompositions across all 29 countries, aggregated country by country. A companion decomposition run on observed intensity covers 26 countries, since Belgium, Cyprus and Lithuania lack a complete matched dwelling-stock series; its envelope term is −11.8 %, those three are absent from that run and present in the one quoted here. Restricting the design decomposition to the same 26 moves the figures to +1.7, +4.7, 0 and −6.3 %.

Replacing the bottom-up demand with the Eurostat residential final-energy reference country by country (available for 24 of the 29 markets, with a strongly non-uniform per-country ratio from about 0.13 to 1.1 and an aggregate near 0.54) leaves every ranking that carries our conclusions unchanged. The base-load cost split holds at 7 markets for the hydrogen boiler and 22 for the best heat pump, the 7 salt-cavern power-peak wins hold, and the four-scenario emissions ordering is preserved. The cost ranking and the cavern set are anchor-invariant by construction, since the levelised costs are computed per MWh of useful heat and the power-peak wins are a short-run-marginal-cost comparison, neither of which carries a demand-volume term. Anchor invariance covers the per-country ordering only. The demand-weighted cost gap is not anchor-invariant, being a weighted mean over exactly the per-country weights the anchor sets, and the base-load split is in any case far more sensitive to the hydrogen price path, the seasonal-storage charge and the network-cost treatment than to the demand anchor (Section S2.7).

For emissions the argument runs through dominance alone, with no appeal to independence. Higher ambition lowers emissions in every market, so one scenario's per-country emissions dominate the next scenario's, country by country, which we confirm pointwise across all 29 markets with no exception at 2050. Any positive per-country reweighting, uniform or not, therefore preserves the ordering.

Recomputing the four 2050 totals under the non-uniform Eurostat ratios returns about 192, 126, 5 and 67 $MtCO_2$ for Current Policies, Stated Policies, Net Zero and H2 Push, the same order with its margins intact. The roughly 9 per cent peaker capital recovery, a per-kW quantity, is likewise unchanged under a proportional rescaling of the firm fleet. Only the extensive levels scale with the anchor, by roughly 0.54; the 2025 baseline of about 644 $MtCO_2$ falls to about 349, and the 278 GW firm fleet and its cumulative capital shortfall fall with it. Because we report the fleet and the shortfall as upper bounds on an inflexible high-demand load, a lower anchor only makes them more conservative.

The United Kingdom, a headline cavern market that lies outside the EU-27 Eurostat series and carries a −28 % vintage-matched deviation of its own, keeps both its base-load ranking and its cavern-market status under either anchor; only its absolute demand and fleet contribution scale. What remains genuinely outstanding is a full decomposition of the Eurostat-versus-Hotmaps 2015 gap (2,075 vs 3,863 TWh) into its prebound/performance-gap, useful-versus-final-energy conversion and stock-coverage components, each with quantified bounds; we leave that decomposition to future work and keep the ranking here.

We benchmark the reconstruction against three independent demand datasets, of which Fig. S1 draws the first: the Hotmaps top-down surface, the EU Building Stock Observatory (BSO) and the Odyssee final-energy series. The bottom-up EU-27+UK+CH total sits −0.8 % from Hotmaps and −11.9 % from the BSO, and lies well above the Odyssee final-energy series ($\approx +53\,\%$); the last of these is consistent with the prebound and performance-gap effect discussed above, not a demand-side error, since Odyssee reports metered final energy and the reconstruction estimates useful heat. We make no claim of agreement with a published-study range for the headline technology shares, for which no matched comparator table exists. The full LMDI derivation, the retrofit depth weights and the per-country validation table go to the public repository (Data availability).

S1.3 Technology and cost layer

We model eight heating technologies: gas, oil and biomass boilers, a resistance heater, air-source and ground-source heat pumps, district heat and a hydrogen boiler. Each is priced on a levelised cost of heat (LCOH) whose denominator is the useful heat the unit delivers over the year. Installed capital is annualised at a country weighted-average cost of capital (WACC) and spread over that

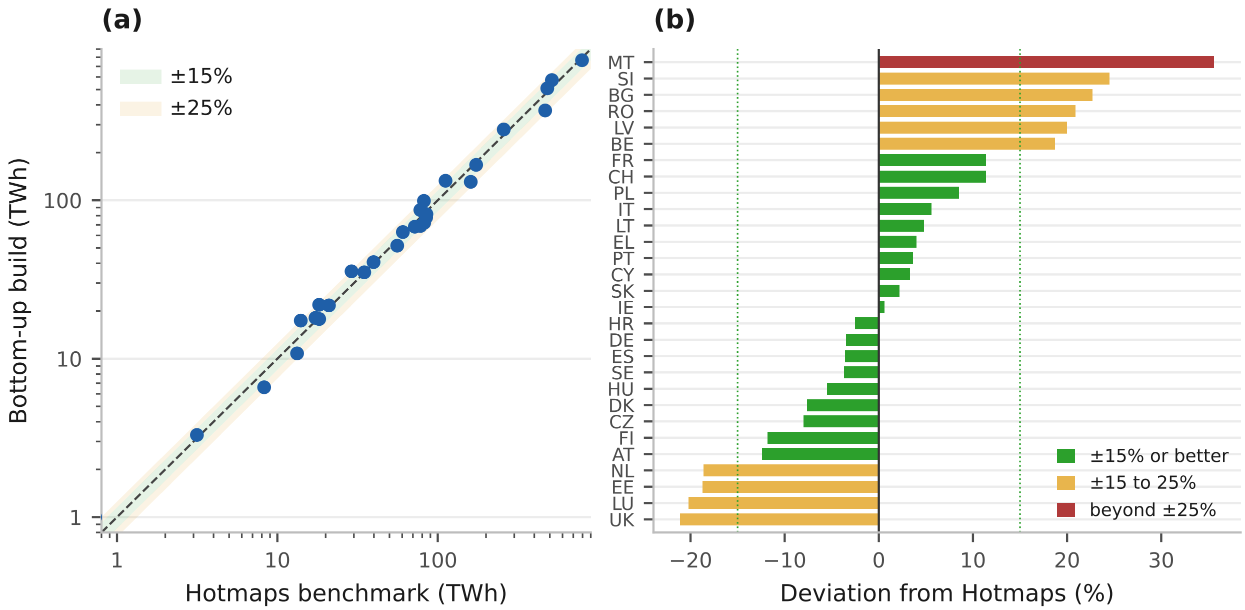

Figure S1: **Bottom-up demand reconstruction against the Hotmaps top-down surface.** Per country (a), on log axes with ±15 and ±25 per cent guide bands, and as the percentage deviation (b).

Note: colour marks countries inside ±15, inside ±25 and beyond. The aggregate lies $-0.8\,\%$ from Hotmaps. The Building Stock Observatory and Odyssee comparisons and the vintage-matched 2015 backcast are given in the text and are not drawn here.

heat output. The country values are our own construction, ordered on published sovereign risk premia [11] and on the cross-country energy-WACC ordering of the renewable-project literature [12], and run from 3.0 per cent in Austria, Luxembourg and Switzerland to 8.0 per cent in Greece. They are not taken from a published heating-sector series, because none covers all 29 markets on a common basis. Recent power-sector estimates span a comparable range on a comparable ordering [13], though for generation assets and not for household equipment. Fixed operating cost is charged the same way. Neither term is divided by the technology's efficiency or seasonal coefficient of performance, because equipment is sized to peak heat demand and a more efficient machine is not thereby a smaller one. Only the fuel and variable operating terms carry the efficiency, since only those scale with the energy the unit consumes. Formally,

$$\mathrm{LCOH}_{c,\tau} = \frac{\mathrm{CRF}(r_c, n_\tau)\, K_\tau + F_\tau}{Q_c} \;+\; \frac{p^{\mathrm{fuel}}_{c,\tau}}{\eta_\tau} \;+\; v_\tau, \tag{S3}$$

where K_τ is installed capital per kW of peak heat output and F_τ fixed operating cost in € per kW per year. F_τ is an absolute charge and not a fraction of capital, so as capital costs fall to 2050 the implied ratio rises, from about 2.1 to 3.1 per cent for an air-source heat pump. Both terms carry a country labour multiplier, on a 70 per cent labour share for F_τ and on a technology-specific share of 20 to 55 per cent for K_τ. Q_c is the annual useful heat delivered per kW of installed capacity, set at 2,000 full-load hours scaled by the country's heating degree days against a 2,800 European reference, so it runs from about 570 h in Malta to 3,710 h in Finland and is applied identically to all eight technologies. $p^{\mathrm{fuel}}_{c,\tau}$ is the delivered fuel price, η_τ the efficiency or seasonal coefficient of performance and v_τ the variable operating cost per MWh of heat. The efficiency appears once, in the fuel term alone. Charging the capital and fixed-operating terms through the efficiency as well would count that efficiency twice, and it would understate a heat pump's capital charge by about €32/MWh at a seasonal coefficient of performance of 3.5 to 4.3, against under €3/MWh for a boiler at an efficiency near 0.9. That distinction is material to the base-load comparison.

Table S3 lists the techno-economic parameters, in EUR_{2024}, with 2025 and 2050 endpoints between which capital cost is linearly interpolated. Values are anchored to the Danish Energy Agency Technology Catalogue installed-cost figures [14], cross-checked against the JRC Technology Data repository [15, 16], the IEA *Future of Heat Pumps* [17] and IRENA [18].

Heat-pump performance enters as a seasonal coefficient of performance (SCOP). The *installed-stock* range is about 2.8 to 4.1 across the existing fleet, on the rated seasonal performance factors EHPA reports by technology and climate zone [19]; low-temperature field performance is measured separately [20]. The SCOP entries in Table S3 (air-source 3.3 to 3.5, ground-source 3.8 to 4.3) are instead *new-install* design values, set on the EHPA/Eurovent rating basis. The air-source entries sit inside the installed-stock field range; only the ground-source 4.3 sits above its 4.1 ceiling, because a design value reflects favourable rating conditions that in-situ operation across an ageing fleet does not reproduce. Because the seasonal coefficient falls in colder climates and at higher flow temperatures, we apply a temperature-dependent country-level derate wherever peak performance is priced (Section S1.6).

The carbon price on fossil fuels is the second emissions trading system for buildings and road transport (ETS2), which launches in 2027 under a price-containment design, an initial soft ceiling near €45/tCO_2 backed by the market stability reserve and additional-allowance release [21]. Our central path starts at €75/tCO_2 in 2027, above that initial ceiling, and follows the BloombergNEF central forecast to €122/tCO_2 by 2030 [22], reaching €200/tCO_2 by 2050; the Monte Carlo low-to-high bracket (€70 to 200/tCO_2 at 2030) sits inside the PRIMES policy range of €71 to 261/tCO_2 [23], so the heat-pump-versus-gas result does not rest on an unusually high fossil carbon price.

The three paths are set out in full in Table S1, because two of the four scenarios run on the high path and the power-arena result is sensitive to it. The high path reaches €350/tCO_2 by 2050, which is what lifts the carbon-priced gas peaker to about €256/MWh-e in Section S1.6. On the central path the gas peaker sits near €181/MWh-e, the cavern advantage narrows from €72 to 89/MWh-e to a few euros, and the seven-market power-arena count is not preserved. The seven should therefore be read as a high-carbon-path result, unlike the base-load ordering, which holds across all three paths at counts of 19, 24 and 29.

Table S1: **Carbon-price paths.** EUR_{2024} per tonne CO_2.

Year	Low	Central	High
2025	30	55	70
2027	45	75	100
2030	70	122	200
2035	100	150	250
2040	130	180	300
2050	150	200	350

Current Policies takes the low path, Stated Policies the central, and Net Zero and H2 Push the high. The 2030 column is anchored on the BloombergNEF central forecast and the PRIMES policy range [22, 23]; the 2035 to 2050 rows are interpolated to a 2050 endpoint consistent with long-run POLES-family projections and are the least well constrained inputs in the cost layer.

That result does sit on the carbon path and moves with it, from 19 of the 29 markets for the best heat pump on the low path, through 24 on the central path we quote throughout, to 29 on the high path. Carbon pricing leaves the heat-pump-versus-hydrogen comparison unmoved at 2050, though not because neither technology pays it. Electricity carries the increment of its carbon cost above the 2025 level, whose median is about €0.7/MWh of useful heat at 2030 on the central path, rising to €5.1/MWh in Poland on the coal-heavy grid, zero at the median from 2040 and zero in every market by 2050. The base-load hydrogen win counts reported in Section S2.2 are therefore identical under the low, central and high carbon paths at 2050.

Capital is additionally scaled by a country construction-labour multiplier. Residential gas and electricity prices for the 2025 baseline are Eurostat first-half-2025 figures [24, 25], and the green-hydrogen hub price follows the European Hydrogen Observatory central trajectory [26, 27] (€200/MWh in 2025 to €50/MWh by 2050).

Retail prices are carried forward from the 2025 baseline by the real-terms multipliers in Table S2, applied to every country's own 2025 level. These multipliers are an assumption of this study rather than a published projection, and they do real work. The 0.55 on gas and the 0.80 on electricity are what take the study-area medians to about €55/MWh and €209/MWh by 2050, and so set the size of the fiscal asymmetry discussed in the main text. A reader who rejects them should read the base-load counts as conditional on them. The hydrogen supply-cost and distribution-infrastructure build are given in the public repository (Data availability).

Table S2: **Real-terms retail fuel-price multipliers.** Relative to each country's own 2025 level.

Year	Gas	Electricity	Hydrogen	Oil	Biomass	District heat
2025	1.00	1.00	1.00	1.00	1.00	1.00
2030	0.90	0.95	0.65	0.90	1.05	0.95
2035	0.80	0.90	0.50	0.80	1.05	0.90
2040	0.70	0.85	0.40	0.70	1.05	0.85
2045	0.60	0.82	0.35	0.60	1.05	0.82
2050	0.55	0.80	0.30	0.55	1.05	0.80

The gas and oil paths assume declining fossil demand across Europe, the electricity path assumes falling generation costs partly offset by network and levy recovery, and biomass rises slightly on competing demand. The hydrogen column is not used for the delivered hydrogen price, which follows the European Hydrogen Observatory trajectory separately. These are among the least well constrained inputs in the study and are swept in the sensitivity analysis.

Delivery-network adders. Two last-mile network charges are applied by default in every result reported here, one to each carrier, so that neither is priced as though its delivery infrastructure were free. The hydrogen boiler carries a distribution charge for the local network that would take the molecule from the transmission backbone to the dwelling. It is computed in the coverage-weighted *blend* mode at its central bound, which converts part of the existing gas distribution network and builds the rest new, weighted by each country's gas-network coverage of its residential stock. The three alternative treatments are free reuse of the existing gas grid, retrofit conversion throughout at the per-property cost reported by the H21 programme [28] and dedicated new-build throughout, and Section S2.7 sweeps all four.

Electric heating carries the matching charge, a low- and medium-voltage reinforcement adder for the winter heat-pump peak, which prices the local distribution capacity that a concentrated electrified heat load requires; across its own bound it adds €1.1 to 38.2/MWh to the heat-pump LCOH. Both adders are on in the baseline, and switching both off recovers the free-infrastructure case, in which neither carrier pays for its own last mile. These two adders, together with the capital-annualisation convention of Eq. (S3) and the seasonal-storage charge below, set the headline base-load result, which is why we document them as baseline assumptions here and sweep them in the extended results.

The missing money is a property of the assumed price cap. The capital-recovery result has a simple closed form, and setting it out here bounds how much weight the result can carry. The peak clearing price is set at the cheaper firm unit's short-run cost plus the scarcity premium π^{s}, so wherever hydrogen is the cheaper unit its inframarginal rent is exactly π^{s} per megawatt-hour and its recovery is $\pi^{\mathrm{s}} \times \mathrm{FLH}/\text{annualised capex}$, with π^{s} in €/MWh converted to €/kW-yr before the division. Germany runs 48 hours, so €60/MWh earns $60 \times 48/1000 = 2.9$ €/kW-yr against €53.4/kW-yr of annualised capital, which is 5.4 per cent; Denmark runs 343 hours, so it earns €20.6/kW-yr against €51.4/kW-yr, which is 40.0 per cent. Two things follow. The storage adder and the delivered hydrogen price both cancel from the rent, so the capital-recovery verdict is invariant to the storage condition that decides the base-load and peaking tests, and it is universal for that reason rather than because of any geology. And the verdict is a direct function of π^{s}, which is an assumption.

That assumption is in tension with another the model makes. The dispatch layer imposes a three-hour sizing standard and values unserved energy at €8,000/MWh, but that value enters only as a penalty inside the unit-commitment objective and is never paid to a generator. Pricing the three shortage hours the model itself imposes at its own value of lost load is worth €24/kW-yr, against an annualised requirement of €51 to 66/kW-yr. Adding it lifts capacity-weighted recovery from 8.7 to about 51 per cent, and Denmark from 40 to 87. The peaker still recovers none of its capital in full, in any of the five markets that build, so the direction survives, but the quantities do not. We therefore report the 9 per cent figure as conditional on an energy-only market with a €60/MWh scarcity margin and no scarcity pricing, which is an assumption about market design and says nothing about the plant itself. Readers should take the finding as the classic missing-money argument [29] applied to a hydrogen peaker, not as evidence that efficient scarcity pricing would leave one unfinanceable.

How much of the result the geology actually carries. The storage charge has two parts, a level and a geological split, and they do not carry equal weight. The level is decisive. Sweeping the charged fraction from zero to the full seasonal adder moves the base-load count from 12 through 22 to 29 of 29 (Section S2.7). The split is not. Repricing every market at the cavern rate of €1.5/kg leaves the heat pump ahead in 19 of 29; repricing every market at the non-cavern €3.0/kg leaves it ahead in 24. The whole geological binary is therefore worth three markets below the adopted 22 and two above it.

Reading the seven country by country makes the point sharper. Charged the non-cavern rate, Denmark still leads by €17.6/MWh, Germany by €13.0, the United Kingdom by €7.8, and Belgium and Ireland are unaffected on €7.5 and €5.8 because they pay that rate already. Only the Netherlands and Poland change hands. Cavern endowment decides two of the seven; what decides the other five is the retail electricity tariff they are compared against. We therefore report the concentration of hydrogen's holds in cavern markets as a co-occurrence and a mechanism that sets the magnitude of the storage term, and we do not claim that geology selects the set. The fiscal asymmetry documented below is the larger cause, and it is why we treat it as the central limitation on the seven.

Two cold-snap coefficients, not one. The dispatch tests derate the heat pump to $\mathrm{COP}_c^{\mathrm{cold}}$, a country-specific figure with a study-area median of 2.04. The electricity reinforcement adder does not use it. It sizes the dwelling's peak draw on a fixed 2.3 across all 29 markets, and the hourly power dispatch uses a third convention, a blended air and ground seasonal coefficient with a load-proportional derate reaching 0.55 at the peak hour and no floor. The three are separate parameters and the paper should not be read as applying one. Substituting the dispatch derate into the reinforcement adder moves its median from €4.9 to about €5.5/MWh and its maximum down from €19.1 to €16.0; the base-load count is unchanged at 22, so this is an internal-consistency gap rather than an error in a reported figure.

What the symmetry does not cover. The "each carrier pays its own last mile" rule holds at the distribution layer of the levelised comparison and is incomplete in three places, all of which we bound here.

The first is that the rule does not travel into the dispatch arenas. In the base-load comparison the hydrogen boiler is charged the blended conversion-and-new-build adder, a study-area median of €13.4/MWh. In the building winter-peak arena of Eq. (S4) it is not. Hydrogen enters as delivered fuel plus the seasonal store, while gas enters at the Eurostat residential retail tariff, which carries network charges and taxes inside it. The asymmetry runs in hydrogen's favour and it is not small. Charging hydrogen the model's own adder in that arena takes the four counts from 0, 4, 9 and 16 of 29 to 0, 2, 7 and 14, and the demand-weighted derived ceiling under Stated Policies from 5.1 to 0.6 per cent. We report the counts on the published basis for continuity with the arena as constructed, and we state here that a fully symmetric dispatch comparison is the lower pair. Together with the scarcity-premium bracket below, this is the second of two independent reasons the Stated Policies ceiling should be read as a range rather than a point.

The second is the electricity side of the seasonal problem. Hydrogen is charged the store its winter-weighted load requires, but electrified heat is charged only distribution reinforcement, leaving out the firm generating capacity that the same winter evenings call on. The model computes that fleet elsewhere in this paper, at about 278 GW costing €51 to 66/kW-yr, or €14.3 to 18.5 bn a year. Two attributions bracket what it is worth on the heat side.

The denominator is the 1,273 TWh of heat-pump heat the 2050 Stated Policies mix carries, which is its 36.5 per cent heat-pump share of 3,483 TWh. Charged pro rata to heating's 9.2 per cent share of power demand, the fleet is €1.0 to 1.3/MWh of heat-pump heat, and the base-load count is unchanged at 22 of 29. Charged in full to the heat-pump load, which treats the whole firm fleet as built for electrified heat, it is €11.2 to 14.5/MWh, and the count falls to 17 and then 15 of 29, with hydrogen picking up Austria, Switzerland, Finland, France and Luxembourg at the low bound and the Czech Republic and Italy in addition at the high one.

Between those two sits the attribution the model can compute directly, and it is the one cost causation asks for, because a peaking asset is built for a peak and not for annual energy. Re-running the dispatch with the heat-pump load removed drops the firm fleet from 277.7 to 56.6 GW, so 221 GW, or 79.6 per cent of the fleet, exists because heat is electrified. That share is what the two dispatch runs support, and we report it rather than a system-peak increment because no committed artefact carries a heat-pump-free system peak. It far exceeds heating's 9.2 per cent of electricity demand, because the cold snap collapses the heat pump's coefficient of performance just as demand peaks. At that share the charge is €8.9 to 11.5/MWh and the count falls to 19 and then 17 of 29, hydrogen picking up Switzerland, France and Austria at the low bound and Luxembourg and Finland in addition at the high one.

We therefore report three attributions rather than two. The pro-rata figure is the lower bound and treats the fleet as serving the whole load; charging it in full is the upper bound and no one would argue it; the incremental figure is the one a cost-causation regulator would use, and it is nearer the upper bound than the lower. The headline of 22 holds only on the lower bound. On the incremental basis it is 19 to 17, and Section 3.2 of the main text reports all three so the count can be read on whichever basis a reader accepts. Note that the incremental share is itself an ordering choice, since it charges electrified heat as the last load added; adding it first would attribute far less. That ambiguity is intrinsic to marginal cost allocation and we do not resolve it here.

The third gap is district heat, which pays no last-mile charge at all while the other two carriers do. Charging it the same €5,000 per connection that this paper's own infrastructure bill already assigns it, annualised over 40 years at the country cost of capital, adds a median €21.0/MWh (range €7.8 to 184, the upper end being the markets with the least heat per dwelling to spread it over), and it moves district heat from cheaper than the best heat pump in 26 of 29 countries to 7 of 29. That charge does not touch the hydrogen-versus-heat-pump comparison this paper leads with, since neither carrier is district heat, but it does bear on the least-cost program's district-heat-led mix, and we flag it as the largest unpriced network asymmetry remaining in the cost layer.

Seasonal storage on base-load hydrogen. The base-load levelised cost also charges hydrogen for the store that its own demand shape requires, on the same basis the peak and dispatch tests use. Space heating is strongly seasonal while electrolytic production is comparatively flat, so a hydrogen heating load creates a winter-to-summer imbalance that has to be held somewhere. On the model's own monthly shape, the share of annual residential heat falling between October and March has a median of 0.748 across the 29 countries and a range of 0.710 to 0.790. The winter share takes three values rather than 29, assigned by a heating-degree-day band, so 19 of the 29 countries, including all seven that hydrogen holds, carry the same 0.248 shift fraction, with 5 above and 5 below. Within the storage term the cavern binary is therefore what separates countries.

Heating is seasonal to that degree while electrolytic supply is far flatter, so we take the fraction of the annual hydrogen heating load that must be time-shifted to be the country's own winter share less one half, which has a median of 0.25. That fraction is charged at the same stored-hydrogen costs used in Eq. (S4), €1.5/kg where domestic salt caverns exist and €3.0/kg where they do not, over the cavern geology mapped by Caglayan and colleagues [30], at about one cycle a year. The base-load charge divides those costs by the year-specific boiler efficiency, 0.92 at 2050, where Eq. (S4) divides by the 0.90 of its own peaking convention, so the resulting adder on the hydrogen boiler's levelised cost is about €12.1/MWh in Germany, a cavern market, against about €28.4/MWh in Spain, which has none.

The cavern binary is a country-level reading of that mapped potential, entered as the fixed list DE, DK, FR, NL, PL, RO and UK rather than derived by applying a working-volume threshold to the maps, and that reading is a judgement of ours. The source also maps potential in Iberia, which we exclude as too small and too far from the demand centres to serve a national heating role, and it maps France's potential in the north alone, which we nonetheless credit to the whole country. The two rates are an assumption of this study rather than a result of it, and the €50 against €100/MWh-heat difference they produce is the largest single term separating the winning markets from the rest.

Two features make this a deliberately conservative charge, and both run in hydrogen's favour. It credits the molecule with meeting the whole non-seasonal half of its load from contemporaneous production, which assumes an electrolyser fleet and a backbone able to follow half the heating load in real time. It also ignores the swing within the winter itself, between a mild December week and a cold snap, which needs a store of its own that we do not price here. The consequence of the charge is set out with the extended results (Section S2.7), where the charged fraction is swept continuously.

Charging nothing leaves the best heat pump cheapest in 12 of the 29 markets; charging half the load-derived shift fraction gives 19; charging the fraction itself, which is the basis adopted here, gives 22; and charging the full seasonal adder used at the peak, about four times the load-derived fraction, gives all 29. An asymmetric accounting, charging the adder in full in the peak and dispatch tests while charging none of it in the base-load test, prices the same molecule two ways and makes the arenas disagree for an accounting reason. Charging it consistently is what makes the base-load and peaking verdicts turn on the same physical quantity.

Fiscal asymmetry in the base-load comparison. The base-load levelised-cost comparison prices electricity and hydrogen on opposite sides of the tax line, and this is the single most important limitation on the seven-market exception it produces. Residential electricity enters at the delivered retail price a household actually pays, a study-area median of about €209/MWh at 2050 once network charges, levies and value-added tax are included. Hydrogen enters at about €50/MWh, a delivered price that carries transport and, through the adder above, last-mile distribution. An independent large-scale assessment of European green-hydrogen supply puts the 2050 marginal cost below €85/MWh [31]. The two are not stated on the same basis, since that figure is a cost of production and ours is a delivered price that already carries transport, so our €50 implies a production cost lower again and sits nearer the optimistic edge of the published range than the arithmetic gap between the two suggests. It carries no excise, no policy levy and no consumption tax, because no such regime exists for a fuel that is not yet sold to households. The gap between the two is therefore in substantial part a gap between fiscal treatments, and hydrogen's residual 2050 base-load lead in seven markets should be read as a statement about storage geology and about how European states currently tax electricity, not as a statement about the two technologies.

We neither model a hydrogen excise regime nor rebate electricity levies, since both would require assumptions about future fiscal policy that no source supports. Two consequences follow. First, equalising the treatment of the two carriers, in either direction, would move the base-load comparison against hydrogen, by an amount we do not size, because the levy-equalised run is not one we perform. Second, that advantage also assumes the molecule is green at about €50/MWh. An Oxford Institute for Energy Studies assessment of the German heating sector concludes that any hydrogen actually supplied to heating would be blue [32], and our own blue route prices that molecule at about €81/MWh [33], a wedge of €31/MWh of fuel that on its own exceeds the €30/MWh by which hydrogen leads in Denmark, its widest surviving win. The blue figure is substituted into the slot the green one occupies, so the repricing is internally consistent, and if the published blue cost is a production rather than a delivered figure the true wedge is wider still, which cuts the same way.

The capacity-side findings are unaffected by the tax point, since the peaking comparison prices both turbines at wholesale and neither pays a household tariff. The provenance point does bear on them, and we separate the two. A dearer molecule raises the hydrogen turbine's short-run cost against an unchanged gas turbine, so we reprice the peaker at the blue figure directly. Read as a European price, €81/MWh in every market lifts the hydrogen turbine to €289/MWh-e against the gas peaker's €256, and the cavern win reverses in all seven. Read instead as a green-to-blue ratio, so that each market's own price scales by 81/50 and keeps its relative position, the seven hold, but their advantage narrows from a band of €72 to 89/MWh-e down to €30 to 57. The single-country estimate does not settle which reading is right, and the paper does not have the supply-side detail to choose, so the peaking win is materially exposed to hydrogen's provenance and may not survive it.

What does survive either reading is the capital-recovery verdict itself. A winning peaker's inframarginal rent is the scarcity premium times its run hours, which carries no fuel-price term at all, so the recovery arithmetic is invariant to the hydrogen price. What the repricing moves is the set of markets that reach that arithmetic in the first place.

Table S3: **Techno-economic parameters by heating technology.** In EUR_{2024}. Capital cost and efficiency are anchored to the Danish Energy Agency Technology Catalogue [14] and cross-checked against [15–18]; seasonal coefficient of performance values follow [19].

Technology	Capex 2025 (€/kW)	Capex 2050 (€/kW)	Eff./SCOP 2025	Eff./SCOP 2050	Fuel
Gas boiler	420	400	0.92	0.94	Natural gas
Oil boiler	450	430	0.90	0.90	Oil
Biomass boiler	950	850	0.88	0.90	Biomass
Resistance heater	200	180	0.98	0.99	Electricity
HP (air-source)	1,200	800	3.3	3.5	Electricity
HP (ground-source)	2,000	1,400	3.8	4.3	Electricity
District heat	500	450	0.95	0.97	District heat
H_2 boiler	600	550	0.90	0.92	Hydrogen

Source: Authors' contribution.

How the technology mix actually moves. The main text describes the four scenarios as setting technology shares, and that is two-thirds true. The interpolation has three parts and we set them out because later sections use the objects by name.

Each share follows $s_t = s_{2025} + (s_{2050} - s_{2025})\,\tau^{p_i}$, with τ the fraction of the 2025 to 2050 window elapsed and a per-technology shape exponent sampled each draw: $p \sim U(0.7, 2.0)$ for heat pumps, $U(0.8, 2.2)$ for district heat and $U(1.0, 2.5)$ for hydrogen, with fossil and resistance reusing the district-heat draw and biomass the heat-pump draw. Because hydrogen's exponent is bounded below at one, a hydrogen draw can back-load its transition but never front-load it. The combined mass of the non-fossil options is then reallocated among them by a softmax on levelised cost, $w_i \propto \exp(-c_i/T)$ at $T = 50$ €/MWh, so a cheaper technology takes a larger share and not the whole market. The weight on that reallocation ramps linearly from zero in 2025 to 0.50 by 2050, and the blended share is the interpolated path and the cost-driven allocation combined at that weight.

A fourth step then applies and it is specific to hydrogen. After the blend, hydrogen is capped at its own drawn 2050 target and any excess is routed to heat pumps up to their feasibility ceiling and then to district heat, after which the mix is renormalised. The cap is one-directional, since cost can push hydrogen below its drawn target but never above it. It exists because hydrogen enters this model as an exogenous adoption setting for a peaking producer rather than a base-load winner, and without it the softmax and the fossil-ban reallocation inflate hydrogen well past its design value; uncapped, the softmax realises about 13 per cent under Stated Policies against a 5 per cent setting. The consequence for reading the results is that hydrogen's realised share is bounded above by construction, so the recurring statement that hydrogen reaches at most about 8.5 per cent of useful heat in any scenario we run is a property of this cap and not an independent finding of the cost mechanism.

Three things follow. By 2050 half the non-fossil allocation is decided by cost, so the mix is not a pure scenario input and the small H2 Push overshoot above the 8 per cent lever comes from the sampling of the lever rather than from cost. The fossil share is governed by the policy-ambition lever throughout and is not reallocated by the softmax, which is why policy ambition alone separates the emissions outcomes. And the temperature T is a judgement we set for plausibility and did not estimate from switching data. It is the largest unestimated parameter on the demand side, and replacing it with a discrete-choice allocation estimated on observed switching is the natural extension.

S1.4 The emissions layer

Emissions are computed after the technology split, as a scope-1 direct inventory over final energy by technology, country and year. Each Monte Carlo draw carries its own inventory, so the fan reported in Section S2.1 is the spread of the sampled mix and not a separate emissions uncertainty.

Combustion fuels carry fixed factors on a lower-heating-value basis, natural gas at 0.202 and heating oil at 0.265 tCO_2/MWh of final energy, from Annex II of the Sixth Assessment Report's Working Group III volume, cross-checked against the UK government conversion factors. Biomass and hydrogen are entered at zero, which is the European accounting convention for a renewable fuel rather than a physical claim; for biomass at 17 per cent of the 2025 mix that convention is material, and it flatters every scenario equally, so it moves the levels and not the ranking. Electricity and district heat carry country-by-year intensities rather than fixed factors. The electricity intensity starts from observed 2025 country values [34] and follows a declining path to 2050 that each scenario scales by its grid-decarbonisation lever, running for example from 600 to 30 gCO_2/kWh in Poland and from 50 to 5 in Switzerland, and district heat follows a parallel country path [35]. Values between the milestone years are linearly interpolated.

Two properties of that construction bound what the emissions results can carry. The post-2025 intensity paths are scenario-consistent trajectories calibrated to published 2025 levels and to European decarbonisation targets, not published country projections, so the absolute 2050 levels inherit that calibration while the ranking across scenarios does not, because every scenario reads the same path through its own lever. And the intensity is an annual average applied to all heating electricity, while the cost layer prices the heat pump's winter draw hourly, which biases the inventory in the heat pump's favour by an amount no hourly ledger here can quantify (Section S2.9).

S1.5 Scenario design

We define four multi-lever scenarios to bracket the plausible range of European buildings decarbonisation to 2050 (Table S4). Each scenario sets six levers: the envelope-renovation rate, the carbon-price path, the hydrogen-price path, the grid-decarbonisation speed, the 2050 fossil-share ambition, and a hydrogen-share ceiling that is bounded by the merit-order dispatch analysis (Section S1.6). **Current Policies** freezes today's implemented measures; **Stated Policies** delivers announced policy, broadly consistent with the IEA World Energy Outlook and the European Commission long-term strategy [36]; **Net Zero** represents climate-neutrality-consistent ambition in line with the IEA Net Zero scenario [37]; and **H2 Push** is the deliberately hydrogen-favourable world of REPowerEU [38], in which hydrogen prices fall along the rapid delivered-hydrogen path [39]. All levers are sampled stochastically around their central values in the Monte Carlo (Section S1.8); per-country lever detail is tabulated in the public repository (Data availability).

S1.6 Merit-order, missing-money and power layer

The weather the power layer runs on. Everything in this subsection is computed on synthetic hourly time series, not on reanalysis or metered data, and on a single realisation of one year. We

Table S4: **Scenario definitions.** Each lever setting and the resulting 2050 demand-change outcome.

Lever	Current Policies	Stated Policies	Net Zero	H2 Push
Envelope rate (vs announced)	$\times 0.70$	$\times 1.00$	$\times 2.73$	$\times 1.64$
Carbon-price path	Low	Central	High	High
Hydrogen-price path	Stranded	Central	Central	Rapid
Grid-decarbonisation speed	$\times 1.15$	$\times 1.00$	$\times 0.70$	$\times 1.00$
Fossil share 2050 (ambition)	40%	27%	5%	15%
H_2 ceiling 2050 (bounded)	0%	5%	3%	8%
Demand change 2050 (outcome)	−5%	−9%	−31%	−18%

Note: envelope-rate and grid-decarbonisation entries are multipliers on the announced-policy (Stated Policies) central case. Every share row is a lever setting rather than a hard cap, and the realised Monte Carlo medians sit a little above their settings because each draw is censored at its admissible interval rather than resampled (Section S1.8).

Source: Authors' contribution.

set this out first because it is the strongest assumption in the paper and it bounds how several of the results below should be read.

Hourly heat demand is a monthly seasonal profile, taken from each country's own load profile, multiplied by a fixed 24-hour shape and then amplified over a single cold snap of six consecutive days in mid-January, with a sharpened morning peak. Solar availability and the non-heat baseline load are deterministic seasonal and diurnal functions. Onshore wind is an AR(1) process around a winter-higher mean, and over the same six days it is forced down to a *Dunkelflaute* at a quarter of its normal level, in all 29 countries simultaneously. The event is therefore perfectly correlated across the study area by construction.

Three consequences follow, and we state them here. First, the three-hour figure we size the firm fleet to is an order statistic on one synthetic year, well short of a loss-of-load expectation over a distribution of weather years. We use "three-hour standard" throughout as shorthand for the sizing rule, which takes the fourth-highest residual hour. Second, the islanded-versus-copper-plate spread we report brackets an event that the model imposes on every country at once, so it measures the consequence of that assumption without resolving the interconnection question empirically. Third, the length of the cold snap could in principle decide how much a shiftable heat-pump load can help, since intra-day pre-heating cannot move load out of an event longer than the shift window. Sweeping that length is a test of the flexibility bound of Table S9 without refining it, and Table S10 reports the outcome; the reduction is nearly invariant to the event length, because the shiftable share binds first. What the sweep does move is the inflexible fleet itself, and we state that band where the fleet is claimed.

Levelised cost compares technologies as base-load suppliers. Hydrogen's most favourable case, however, is as a peaking producer serving only the cold-snap slice of the load-duration curve, the $\approx$12 to 18 % of annual heat above the 2000-hour demand level, when air-source heat-pump performance collapses and the power system is stressed. We therefore model the winter peak as a merit-order dispatch among three quick-start options, whose marginal costs per MWh of useful heat are

$$c_c^{\text{gas}} = \frac{p_c^{\text{gas}} + \varphi_y \, e_{\text{g}}}{\eta_{\text{g}}}, \qquad c_c^{\text{H}_2} = \frac{p_c^{\text{H}_2}}{\eta_{\text{h}}} + \sigma_c, \qquad c_c^{\text{HP}} = \frac{\pi_c^{\text{peak}}}{\text{COP}_c^{\text{cold}}}, \tag{S4}$$

where φ_y is the carbon price, $e_{\mathrm{g}} = 0.202$ $\mathrm{tCO_2/MWh}$ the gas emission factor, η_{g}=0.92 and η_{h}=0.90 the boiler efficiencies, σ_c the seasonal-storage adder (≈€50/MWh-heat with domestic salt caverns, ≈€100 without, from an assumed €1.5 and €3.0/kg of stored hydrogen at about one cycle a year, over the cavern geology of 30), and $\mathrm{COP}_c^{\mathrm{cold}} = \max(0.60\,\mathrm{SCOP}_c, 1.8)$ the country-level cold-snap heat-pump coefficient of performance, calibrated to the low-temperature field measurements [20]. We do not assume the scarcity electricity price π_c^{peak} that prices the heat-pump branch; we derive it from the power sector's own merit order. The marginal winter unit, a gas open-cycle turbine (OCGT) competing against a hydrogen-ready turbine at backbone fuel prices, sets the price, to which we add a scarcity premium:

$$\pi_c^{\mathrm{peak}} = \min\left(\frac{p^{\mathrm{gas,w}} + \varphi_y\, e_{\mathrm{g}}}{\eta_{\mathrm{OCGT}}},\ \frac{p_c^{\mathrm{H_2,bb}}}{\eta_{\mathrm{T}}} + \sigma_c^e\right) + \pi^{\mathrm{s}}, \tag{S5}$$

with $\eta_{\mathrm{OCGT}} = \eta_{\mathrm{T}} = 0.40$. Both turbines carry the same efficiency in this equation, so the win counts and the €72 to 89/MWh-e cavern advantage rest on an even-handed comparison and cannot be an artefact of the efficiency assumption. The capital-recovery layer is the one that lets efficiencies evolve, taking the gas turbine to 0.42 by 2050 and the hydrogen turbine to 0.44, or 0.48 under H2 Push, whose premise is an advanced simple-cycle machine. That layer therefore hands hydrogen the better machine, and does so deliberately. The missing-money verdict should have to survive a hydrogen turbine more efficient than its fossil rival, and should not rest on penalising one. Holding both at 0.40 there narrows that layer's own cavern advantage from €85 to 99/MWh-e down to €67 to 84/MWh-e and leaves the recovery verdict unchanged, since recovery stays far below the 100 per cent that private financeability requires.

We bracket the scarcity premium as $\pi^{\mathrm{s}} \in \{30, 60, 120\}$ €/MWh (low, central and high), and the recovery sensitivity in the main text evaluates the same bracket on a four-point grid that adds €90. These three are stylised values we choose to span a deeply interconnected, a typical continental and an islanded system; they are not estimated from observed scarcity rents in any market, and we set them as a bracket precisely because no single value is defensible across 29 systems. Where the premium enters decides how much rests on it, and it enters the three arenas differently.

It cancels exactly from the power-arena comparison, as shown below, because there the premium is added to whichever firm unit is marginal and the gas and hydrogen turbines are compared on short-run marginal cost alone. The power-arena count of seven is therefore invariant across the whole €30 to 120/MWh-e band. It does not cancel from the building winter peak, because it is precisely this scarcity price that the cold-snap heat pump is charged, at $\pi_c^{\mathrm{peak}}/\mathrm{COP}_c^{\mathrm{cold}}$. A higher premium widens hydrogen's niche there and a lower one narrows it. Across the bracket the building-peak count runs 0, 1, 8 and 15 of 29 across the four scenarios at €30/MWh, 0, 4, 9 and 16 at the central €60 and 0, 5, 10 and 18 at €120. Every sweep here moves one lever and holds the rest at their published settings, so these run from the asymmetric accounting and the central column is its 0, 4, 9 and 16, not the symmetric 0, 2, 7 and 14 the main text headlines; the last-mile charge is the separate sweep below. The demand-weighted derived ceiling moves with it, most sharply under Stated Policies, where it falls from 5.1 to 0.2 per cent at the low end. That scenario's 5 per cent hydrogen ceiling therefore sits at or below its dispatch-supported bound at the central and high premium but above it at the low one, and we report the bound as a range rather than a point for that reason. A third sensitivity belongs beside those two and had not been swept before. The cold-snap derate $\mathrm{COP}_c^{\mathrm{cold}} = \max(f\,\mathrm{SCOP}_c, \mathrm{floor})$ is published at $f = 0.60$ with a 1.8 floor, a study-area median of 2.04, and the low-temperature field evidence supports roughly 0.5 to 0.75. Across that range the building-peak counts run 0, 5, 10 and 18 of 29 at $f = 0.50$, 0, 4, 9 and 16 at the published 0.60, and 0, 1, 7 and 10 at 0.75, with the Stated Policies ceiling moving from 5.3 to 0.2 per cent. A better heat pump in the cold snap narrows hydrogen's peaking niche, which is the expected direction, and this is the largest single lever on that arena. Taken with the premium bracket and the dispatch-arena last mile, the Stated Policies ceiling is bounded by three independent choices and we report it as a range for that reason.

Nor does the premium cancel from capital recovery, where a winner's rent is the premium times its run hours, so recovery is exactly proportional to it. We compute recovery at the central €60/MWh and every recovery figure in this paper is on that value.

The proportionality means the bracket needs no separate run; capacity-weighted hydrogen recovery across the cavern markets is 8.7 per cent at the central premium and therefore 4.4 and 17.4 per cent at the low and high ends. The missing-money conclusion is the same at all three, since none of them reaches the 100 per cent that unaided financeability requires, and the high end would need a scarcity premium near €690/MWh sustained over the same run hours to get there. The missing-money framing follows [29]. Hydrogen's heating share in each scenario is the peak slice it wins under Eq. (S4), so the ceiling is bounded by the dispatch. It is not forced to equal that bound. Net Zero is held at 3 per cent against a supported 7.9 and H2 Push at 8 against 9.3, because a scenario's wider electrification ambition can displace hydrogen for reasons the peak dispatch does not price.

Throughout, a country counts as a hydrogen *win* in the power arena where its hydrogen-ready turbine undercuts the gas OCGT on short-run marginal cost in Eq. (S5), i.e. $p_c^{\mathrm{H_2,bb}}/\eta_{\mathrm{T}} + \sigma_c^e < (p^{\mathrm{gas,w}} + \varphi_y\, e_{\mathrm{g}})/\eta_{\mathrm{OCGT}}$. We adopt this cost-undercut criterion consistently in preference to a profit- or clearing-price metric, which is more sensitive to the assumed scarcity premium, so every reported power-arena win count rests on one definition. The reported results carry a second, looser short-run cost series alongside this one, used for the peaker profit and capital-recovery accounting; on that series the hydrogen turbine undercuts gas in 15 markets, against seven on the equal series. Every win count, figure and capacity share reported here uses the endogenous winter-peak series above. The seven salt-cavern markets are the subset the equal-efficiency series returns; the looser series is not confined to cavern geology.

That 15 is contingent on one convention. The looser series prices hydrogen for a turbine at 0.85 of its delivered price, on the reasoning that a turbine sits on the transmission backbone and pays no distribution margin. Whether that discount is right turns on whether the delivered price series carries a distribution margin in the first place, and our own cost layer treats it as though it does not, adding the last-mile network separately. Removing the discount takes this count from 15 to 8. Nothing else moves. The equal-efficiency count of seven, the seven-market cavern set, the 8.7 per cent capacity-weighted recovery and every capacity payment are all invariant, because where hydrogen is the marginal unit the clearing price is its own short-run cost plus the scarcity premium, so the rent it earns does not depend on the level of the fuel price. The reader should therefore read the 15 as the upper end of an 8-to-15 range on a convention we have not settled, and the seven as the number the paper's conclusions rest on.

The seven-market cavern concentration and these win counts turn on two exogenous stylised values (the scarcity premium π^{s} and the storage adder σ); the ≈€50/MWh-heat geological difference between salt-cavern and non-cavern markets is the single largest driver of the cavern gap. Of that pair only the storage adder can move the win count. The scarcity premium enters the clearing price additively and outside the minimum in Eq. (S5), so it cancels from the hydrogen-versus-gas comparison exactly; its invariance is an algebraic identity, which no empirical robustness result establishes.

The sensitivity we report in the extended results (Table S14) is therefore a one-dimensional sweep of the storage adder σ across nine values, from €25 to 100/MWh-heat, computed at each of four scarcity premia from €30 to 120/MWh-e. The count is identical at all four, so the table reports one column and states the invariance. The win count holds at seven across storage adders of €50 to 75/MWh-heat and the winning set is exactly the salt-cavern markets over that range; at €45 an eighth market enters, at €38 and below the adder no longer separates the geologies and all 29 win, at €88 only two survive and at €100 none does. The result is therefore a feature of the central half of the swept range only. The sweep scales the cavern and non-cavern adders in lockstep, so it varies the level of the storage penalty and holds the cavern-to-non-cavern ratio at 2.0 throughout; it is a test of the level alone, leaving untested the geological spread that drives the result.

Winning the dispatch is necessary but not sufficient; an investor must also recover the peaker's capital from the hours it runs. The hydrogen peaker earns an inframarginal rent against the cheapest alternative over the peak slice, whose width $H_c \approx 518$ h is the equivalent full-load hours of that slice, its annual energy divided by its capacity, both read from the load-duration curve. Germany is representative at 97.7 TWh over 188.8 GW, or 518 h; across the study area H_c runs from 514 to 534 h with a median of 518 on the unrounded slice fraction; the committed column prints 519 and 535 because it carries that fraction to three decimals. This is the utilisation of the peak plant, distinct from the count of hours above the threshold, which is 2,000 h by construction.

Two hour figures appear in this analysis and describe different quantities, so we distinguish them explicitly to avoid confusion; $H_c \approx 518$ h in Eq. (S6) and the Nomenclature is the *definitional width of the peak slice* (the run window over which the boiler serves cold-snap heat). The ≈139 h quoted in the results is the *realised full-load utilisation* of the power-system hydrogen peaker, which fires at full load only in the subset of those slice hours where it is actually dispatched. Equation (S6) is a heating-technology capital-recovery test (the operating saving of the H_2 boiler against the next-best heat option over its run hours), which no power-market scarcity rent enters, so hydrogen is not treated as simultaneously marginal (Eq. (S4)) and inframarginal in the same market. The annual margin per kW and the build condition are

$$R_c = \left[\min\left(c_c^{\text{gas}},\, c_c^{\text{HP}}\right) - c_c^{\text{H}_2}\right] \cdot \frac{H_c}{1000}, \qquad \text{build iff } R_c \geq \left[\text{CRF}(r_c, n) + m\right] K, \tag{S6}$$

where $K =$ €600/kW is the hydrogen-boiler capital cost, n=20 years, m=2 % fixed operating cost, r_c the country weighted-average cost of capital and CRF the capital-recovery factor. To trace the electrified heat load into the power system, we resolve the dispatchable fleet's role in a capacity versus energy accounting, defining hydrogen's share of the system peak and of generated electricity as

$$\Phi_{\text{cap}} = \frac{\sum_c K_c^{\text{pk}}}{D^{\text{pk}}}, \qquad \Phi_{\text{en}} = \frac{\sum_c E_c^{\text{pk}}}{D^{\text{tot}}}, \tag{S7}$$

where K_c^{pk} and E_c^{pk} are the firm peaking fleet's capacity and energy, which is every quick-start unit the sizing rule builds and not the hydrogen units alone, D^{pk} the system peak and D^{tot} total demand. The reduced-form peak-capacity cap (the closed-form dispatch limit, which the unit-commitment run below supersedes), and the network-capital bill are set out below and tabulated per country in the public repository (Data availability); the full symbol list is in the main-text nomenclature.

The peaker economics are a reduced-form, single-use screening. The model does not co-optimise demand and supply, endogenise storage sizing or demand-side flexibility, or run a capacity-expansion layer, and it prices only the peaker's own-market rent, leaving out the cross-sector firm-capacity value a hydrogen turbine might capture if its capacity were shared with industry and transport. The headline ∼9% capital-recovery verdict is therefore, in part, definitional for a low-utilisation peaker and conditional on the assumed absence of a capacity payment. This makes the missing-money conclusion a robust direction (no private investor recovers the capital of a ∼139-full-load-hour peaker at these premia) but a magnitude that a capacity market or a flexible, peak-flattening heat load could move.

We quantify that payment directly from the reported peaker economics. Net of the recovered rent, a standing payment of about €31 to 63/kW-yr closes the gap for the winning hydrogen peakers, and a gas peaker in the same markets needs about €41 to 52/kW-yr, which separates the definitional low-utilisation effect from the policy-relevant magnitude and attaches the payment to low-utilisation firm capacity rather than to hydrogen. One extension remains beyond the reduced-form screening adopted here. A capacity-expansion co-optimisation with endogenous storage sizing, together with a demand-side/building-thermal-storage case that flattens the winter peak and re-sizes the firm fleet, would bound how far the capacity-not-energy framing and the €357 bn cumulative shortfall depend on the present no-flexibility, single-use assumption.

The generation stack the dispatch runs against. The dispatch runs on a per-country installed-capacity stack for 2025, 2030, 2040 and 2050 that we curate rather than take from one source, because no published per-country 2050 series covers all 29 markets on a common basis. The largest systems are anchored to national 2050 strategies; the remainder take each country's updated national energy and climate plan for 2030 and escalate it on ENTSO-E Ten-Year Network Development Plan National Trends growth. Every row carries a tag recording which of the three it is, and the aggregate is cross-checked against the same plan's totals. The 889 GW system peak, the 278 GW peaking increment, the 195 GW firm combined-cycle fleet and the 0.86 per cent energy share are all conditional on this stack, which is released with the model for a reader who wishes to replace it.

The electricity demand the peaker is sized against. Two denominators carry the capacity and energy shares, and both rest on one assumption. Non-heat national electricity demand is set at its 2023 level, 2,926 TWh across the 29 markets on ENTSO-E and Ember reporting, and grown by a smooth factor reaching 1.40 by 2050 (1.00, 1.10 and 1.25 at 2025, 2030 and 2040) to stand for the electrification of road transport and industry. That is an assumed path, equivalent to 1.35 per cent a year compounding, and no model produces it, and we adopt a single factor across countries because the comparison at issue is between two firm technologies inside one demand envelope. Heat-pump electricity is added on top of it from the heating model, at 417 TWh in 2050, giving the 4,513 TWh total against which the peaker's 0.86 per cent energy share is measured. Both shares scale inversely with this factor. At a flat baseline the peaker's energy share would rise to about 1.2 per cent and leave the capacity share materially unchanged, because the winter peak sets that share and annual energy does not.

Power-system coupling extensions. We strengthen the power-sector coupling in three ways, each derived below. First, an explicit *unit-commitment* optimisation cross-checks the reduced-form peak-capacity cap for the dispatchable fleet. For each country and milestone year we minimise weighted fuel, carbon, variable and start-up cost, splitting the fleet between a firm combined-cycle class and fast open-cycle gas or hydrogen peakers. The dispatch respects minimum stable output, ramp limits, minimum up and down times, a 5 % operating-reserve margin, and battery and reservoir state of charge, and unserved energy is penalised at a value of lost load of €8,000/MWh. That figure is a stylised value we set, not a national estimate, and we make no claim that it matches the values European regulators adopt in adequacy assessment.

Its role in the optimisation is to make load shedding expensive enough that the solver never sheds by preference, so the commitment result is insensitive to its level provided it exceeds every generator's short-run cost, which €8,000/MWh does by two orders of magnitude. It is not insensitive to it where the value is used as a price, and the one place we do that, the counterfactual scarcity-pricing calculation of Section S2.4, is reported as conditional on it for that reason. The commitment problem is solved as a linear relaxation whose integrality gap against the mixed-integer formulation is only 0.04 to 0.11 % on the five countries spot-checked over a seven-day design week, namely Germany, France, Poland, Belgium and Finland. A constraints-off limit test reproduces the reduced-form peak-capacity cap to within 0.40 % per country, confirming the two layers agree where they should.

Second, a *myopic rolling-horizon* foresight test relaxes the perfect-foresight assumption of the least-cost linear program. Re-optimising on a one- or two-period look-ahead leaves the 2050 mix ($\sim$53 % district heat, $\sim$45 % heat pumps, 0.3 % hydrogen) and the present-value system cost (approximately €7,430 bn) unchanged from the perfect-foresight optimum, so neither the near-zero hydrogen share nor the retrofit arm's 2050 zero is a foresight artefact.

Third, an *endogenous Switzerland* treatment closes three couplings that a one-directional exogenous check leaves open. The Swiss delivered-hydrogen cost is made EU-supply-consistent with a 600 km pipeline adder. A fixed-point feedback then couples incremental Swiss heat-pump load to the EU winter electricity price, converging to a tolerance of €0.01/MWh, with the winter price moving once from €240 to €246/MWh-e in 2050. Finally, a seasonal-hydro credit recognises the Swiss reservoir fleet as firm winter capacity. Switzerland is only $\approx$2.3 % of the coupled demand and is interconnector-bound, and the feedback reinforces the heat-pump verdict.

The endogenous heating-to-power feedback is, however, closed at full resolution only for Switzerland. Extending the fixed-point coupling to the full EU aggregate, or bounding the feedback magnitude analytically, would establish whether the EU-wide derived scarcity price is as stable as the Swiss case (where the winter price moves only €240$\rightarrow$246/MWh-e) implies. That reading holds only for the uncoupled arm. Crediting the Swiss reservoir fleet as firm winter capacity in the full endogenous run moves the same price to about €141/MWh-e, so the Swiss case bounds the feedback without showing it to be small. This generalisation is left to future work, and we rely for now on the small size and interconnector-bound position of Switzerland to argue that the EU-wide feedback is of the same limited order.

S1.7 The least-cost program

The cost-optimal cross-check of Section 3.5 of the main text is a linear program, and its formulation is as follows.

Let c index the 29 countries, t the eight heating technologies and $y \in \{2025, 2030, 2040, 2050\}$ the milestone years. The decision variable $x_{c,t,y} \in [0, 1]$ is the share of country c's useful heat met by technology t in year y. The program minimises discounted, period-weighted system heating cost,

$$\min_x \sum_c \sum_t \sum_y \frac{w_y}{(1+r)^{y-2025}} \, \mathrm{LCOH}_{c,t,y} \, D_{c,y} \, x_{c,t,y}, \tag{S8}$$

where, in Eq. (S8), w_y is the number of years the milestone y represents, r the discount rate, $\mathrm{LCOH}_{c,t,y}$ the source-audited levelised cost of heat with the ETS2 carbon price embedded, and $D_{c,y}$ the country's useful-heat demand, held at the Stated Policies LMDI trajectory so the program answers which technology is cost-optimal at a given demand rather than co-optimising demand and supply.

Six constraint families apply. First, shares balance, $\sum_t x_{c,t,y} = 1$ for every country and year. Second, in trajectory mode the first year is anchored to the observed 2025 mix, $x_{c,t,2025} = m_{c,t}$.

Third, deployment ceilings cap district heat, ground-source heat pumps, the two heat pumps together, hydrogen and biomass at each country's feasible share, and a fossil ceiling implements the announced replacement bans. Fourth, stock turnover bounds the change between consecutive milestones at each country's own replacement rate, $|x_{c,t,y_i} - x_{c,t,y_{i-1}}| \leq \tau_c\,(y_i - y_{i-1})$. Fifth, each country must meet a scope-1 emissions cap in every year against *its own* 2025 baseline, with no cross-border pooling,

$$\sum_t e^{\mathrm{s1}}_{c,t,y}\, D_{c,y}\, x_{c,t,y} \;\leq\; (1 - \kappa\,\theta_y)\, E^{\mathrm{s1}}_{c,2025}, \qquad \theta_y = \tfrac{y-2025}{25}, \tag{S9}$$

for ambition $\kappa \in \{0.75, 0.90, 1.00\}$. The cap glides linearly from the country's own 2025 baseline to $(1-\kappa)$ of it at 2050 rather than binding at $(1-\kappa)$ throughout; imposing the terminal cap in every year is infeasible at $\kappa = 0.90$ given the turnover limit. Sixth, all shares are non-negative. The program is solved with CBC.

The dual of Equation (S9) is the implied carbon price country c requires in year y to meet its target, and it is the quantity the main text reports as a shadow price. The raw dual is in the objective's units, which are period-weighted and discounted to 2025, so we divide by $w_y/(1+r)^{y-2025}$ to return it to euros per tonne in year-y money. Reporting the raw dual instead inflates it by that factor, about threefold at 2050.

A second arm of the same program adds envelope retrofit as a decision, so the technology mix reported above is not being asked to carry the whole burden by construction. It introduces $x^{\mathrm{ret}}_{c,y} \in [0, \bar{x}^{\mathrm{ret}}_c]$, the share of country c's stock retrofitted by year y, charges it at the country's annualised retrofit cost and credits the demand it removes, subject to the same turnover limit. The arm relieves the cap slightly rather than transforming the answer. Retrofit takes 2.8 per cent of the stock at 2030 and 5.8 per cent at 2040, falls back to zero at 2050 once the technology mix alone clears the cap, and lowers the present-value system cost from €7,428.4 bn to €7,403.6 bn at $\kappa = 0.90$. The 2050 mix and the near-zero hydrogen share are unchanged, which is why the fixed-envelope arm is the one reported throughout.

Two limits carry from the formulation to the result. The cap is scope-1 only, so electrified heat pays nothing inside the constraint for the power-sector emissions it causes, which mechanically favours the electric technologies and district heat in exactly the direction the reported mix runs. Re-imposing the cap on total rather than scope-1 emissions bounds that bias, and it is small. The 2050 mix moves from 52.9 per cent district heat, 45.4 heat pumps and 0.3 hydrogen to 52.9, 45.6 and 0.3, the present value from €7,428.4 to 7,429.0 bn and the highest binding shadow price from €81 to 83/tCO_2. The direction of the bias is as stated; its size does not carry the result. And the cap is national while ETS2 is a pooled price, so the agreement between the two is about direction and not about mechanism.

S1.8 Uncertainty and sensitivity

We characterise parametric uncertainty with a Monte Carlo of $N = 200$ draws per scenario (a fixed random seed), sampling the per-country envelope-renovation rate $r_{c,s}$, the 2050 technology-share targets, the heat-pump ground/air split, and the per-carrier fuel-price multipliers (log-normal, $\sigma = 0.15$). Because renovation pace is driven partly by common EU policy and partly by national factors, the per-country rate shocks are correlated through a shared factor, $z_c = \sqrt{\rho}\, z_{\mathrm{EU}} + \sqrt{1-\rho}\, z^{\mathrm{idio}}_c$ with $\rho = 0.5$; this preserves each country's marginal distribution (so per-country bands and all medians are unchanged) while widening the EU-aggregate band by about 2.2 times relative to independent draws. The same construction is applied to the per-carrier fuel-price multipliers at $\rho = 0.75$.

Two further points about the correlation structure bear on how the reported bands should be read. First, the amplification is close to a closed form, $\sqrt{1 - \rho + \rho\, n_{\mathrm{eff}}}$ with an effective country count $n_{\text{eff}} = (\sum_c s_c)^2 / \sum_c s_c^2 = 9.2$ on demand shares, which returns 2.26 at $\rho = 0.5$ against the simulated 2.2; the width of the EU demand band is therefore largely a function of ρ, which we set by judgement without estimating it; the band would be 1.7 or 2.7 times as wide at $\rho = 0.25$ or 0.75. Second, correlating these two families is the less conservative of the two treatments. Every remaining sampled quantity, including the 2050 target shares, the uptake exponents, the ETS2 pass-through, the discount rate and the heat-pump-feasibility multipliers, is a single scalar drawn once per draw and applied to all 29 countries, which is perfect cross-country correlation. The two parameters we set below one are the only ones that are not at $\rho = 1$.

Each 2050 target share is drawn about its tabulated setting (Table S4) with a standard deviation of 5 percentage points on the heat-pump share, 4 on the fossil share and 3 each on the district-heat and hydrogen shares. Where a sampled quantity is bounded we censor at the bound without resampling, so mass accumulates on the bound itself. This applies in four places and binds in three and is material in one. Under Current Policies the 2050 hydrogen target is drawn $\mathcal{N}(0, 0.03)$ and censored at zero, which places half the draws exactly at zero and raises the mean of the sampled lever to 0.012, so that scenario realises a median hydrogen share of 0.49 per cent rather than the zero its lever names. A deterministic zero would remove it, at the cost of making one lever behave unlike the other three, and we report the censored draw rather than re-running to a tidier number. The renovation-rate bound is never reached, and the remaining two bind immaterially.

We report the 10th, 50th and 90th percentiles across draws. A running-quantile test on the headline 2050 CO_2 metric (Fig. S2) shows the median settling inside a $\pm 1\,\%$ tube after 41, 48 and 96 draws under Current Policies, Stated Policies and Net Zero, whose response is bimodal (Section S2.1), and only after 139 under H2 Push, the slowest of the four. A running quantile can re-enter the tube and leave it again, so each figure is the last draw at which the series is still outside, which is the conservative reading of the two. The deciles are slower still and are not converged at $N = 200$, since the running 10th percentile last leaves that tube at draw 156 to 188 in all four scenarios. This is a property of the estimator and not of the run. At $N = 200$ the reported 10th percentile is the 20th order statistic, whose exact nonparametric 95 per cent interval spans the 12th to the 29th order statistic, so it pins the population decile only to somewhere between the 6th and the 15th percentile. We therefore read the medians as converged and treat the reported deciles as an indicative spread, not as precisely located quantiles.

We complement the one-at-a-time sensitivity sweeps with a variance-based (Sobol) global sensitivity analysis of the four uncertain *demand drivers* only, the envelope-renovation rate, occupancy (household size), dwelling size and population (Fig. S3). The decomposition is of the variance of EU useful heat demand in 2050 across these four drivers; it is not a variance decomposition of CO_2 or system cost. Because 2050 useful demand depends analytically on these drivers alone, a full Saltelli design with $N = 2{,}048$ is exact and immediate. The four drivers are sampled over deliberately unequal bands, each set to that driver's own plausible range, not to a common width: the envelope-renovation rate over $\pm 35\,\%$, population over $\pm 3\,\%$, occupancy over $\pm 5\,\%$ and average dwelling size over -2 to $+5\,\%$.

The total-order indices are $S_T = 0.40$ (envelope rate), 0.33 (occupancy), 0.16 (dwelling size) and 0.12 (population); first-order indices approximately equal the total-order indices, indicating a near-additive response over these ranges. Because the deterministic projection of Eq. (S1) holds average dwelling size flat, the Sobol screen deliberately relaxes that base assumption and lets dwelling size vary over -2 to $+5\,\%$ purely to probe sensitivity; its $S_T = 0.16$ therefore measures how much 2050 demand would vary over that narrow range, which is distinct from the variance the base projection actually generates. The ranking these indices produce is a joint property of the model response and of those band widths, and it is not invariant to them; the envelope rate leads in part because its band is the widest. Read the indices as the variance each driver contributes over its own stated range, not as a scale-free ordering of driver importance.

The demand-driver Sobol screen covers only the parameters that enter 2050 useful heat demand analytically, and not the parameters that carry the hydrogen, cost and CO_2 headlines (the carbon-price, hydrogen-price, discount-rate/WACC, ETS2-pass-through, fuel-price and technology-share draws, which act instead through the levelised-cost layer). To place these on an equally global footing we add a variance-based screen over the full sampled parameter space. Because the stored Monte Carlo persists only the q10/q50/q90 quantiles and uses independent (non-Saltelli) draws, a Saltelli-based Sobol decomposition cannot be reconstructed from the archived output; we therefore compute standardised rank-regression coefficients (SRRC), a legitimate variance-based global measure that is valid on random (non-Saltelli) designs.

We re-run the real NUTS3 Monte Carlo (the full simulation) for the H2 Push scenario as a separate ensemble of $N = 192$ draws, which is this screen's own sample and not a subset of the 200 the headline results use, and regress the rank-transformed 2050 buildings CO_2 output on the rank-transformed inputs across all $\approx$17 sampled parameters: the fossil, heat-pump, district-heat and hydrogen 2050 target shares, the demand-reduction and demand-shape parameters, the technology uptake-shape exponents, the ground-source share, the ETS2 pass-through, the discount rate/WACC, the heat-pump-feasibility multipliers, and the per-carrier gas, electricity and hydrogen fuel-price multipliers (Table S13). The rank regression explains the output almost completely (rank-$R^2 = 0.95$), and a single lever dominates: the 2050 fossil-phaseout ambition, at SRRC = 0.98. Every economic-price axis is negligible by comparison, with hydrogen price $|\text{SRRC}| = 0.009$, electricity price 0.015, gas price -0.007, ETS2 pass-through 0.011 and the discount rate 0.002 (all $|\text{SRRC}| < 0.02$).

Four qualifications govern how far that result can be pushed, and we state them because the screen is easy to over-read. First, the near-zero price coefficients are not an empirical finding about the world but a consequence of the model's own structure. At 2050 the technology-share interpolation has reached its target exactly and fossil technologies are excluded from the levelised-cost softmax, so the emitting share is a deterministic function of the fossil lever alone. Scaling every fuel price in every country by 0.6 or by 1.6 leaves the 2050 gas-plus-oil share identical to four decimal places, as does moving the discount rate across the full 3 to 8 per cent range. The correct reading is that emissions are price-inert *by construction*. That is a modelling choice, and it is not evidence of robustness.

Second, the rank-R^2 is not independent information, because the squared coefficients sum to 0.955 against an R^2 of 0.95, and the fossil lever alone accounts for 99.8 % of that, so the two statistics restate each other. The 0.95 is also the best of the four. The same regression returns 0.930, 0.942 and 0.432 for Current Policies, Stated Policies and Net Zero, and the Net Zero collapse is its two-branch emissions mixture defeating a rank regression, so the screen is least informative in the scenario the fossil lever has already driven to zero.

Third, the carbon price is not among the sampled parameters; it is a fixed per-scenario trajectory, so this screen says nothing about the carbon axis, and the sampled price and discount bands are far narrower than the scenario spans they might be mistaken for. The hydrogen multiplier spans about 0.87 to 1.19 at the 10th and 90th percentiles, which is roughly €47 to 65/MWh delivered in Germany against a scenario span of €31.6 (Rapid) to €130.8 (Stranded), and the sampled discount rate spans 4.3 to 5.6 % against a country range of 3 to 8 per cent.

Fourth, we run the screen on H2 Push and it does not generalise to all four scenarios. Repeating it identically gives rank-$R^2 = 0.930$ and 0.942 under Current Policies and Stated Policies, but 0.432 under Net Zero, where the fossil coefficient falls to 0.583 because three-quarters of the draws collapse onto the phaseout branch and the response becomes a step and ceases to be monotone. A rank regression is not a valid summary there, and we do not claim one. Taken together the screen supports a narrow statement, that within the sampled economic ranges the 2050 emissions ordering is set by policy ambition, short of the broader claim that the verdict has been tested across the full cost, carbon, hydrogen-price and discount space.

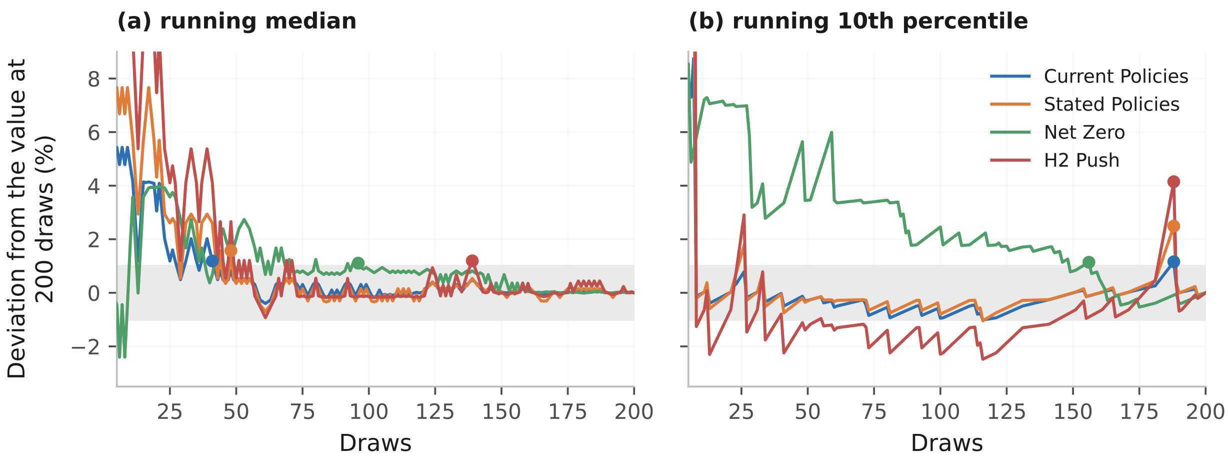

Figure S2: **Running-quantile convergence of the 2050 CO_2 metric.** Each series is the quantile recomputed on the first k draws, as a deviation from its value at $N = 200$. The shaded band is the $\pm 1\,\%$ tube and the marker is the last draw outside it.

Note: the first few dozen draws overshoot to $+27\,\%$ in panel (b) and are clipped, so the axis resolves the settled tail the convergence claim is about. Medians settle; the deciles are still leaving the tube at draw 156 to 188, which is a property of the order statistic at $N = 200$ rather than of this run.

S1.9 Reproducibility and data availability

The model, the scenario configurations and the analysis scripts are released in a public repository under an MIT licence, with download scripts for the Hotmaps, Eurostat and GISCO source datasets, the UK ONS TS044 accommodation-type table being a manual one-time download that is committed to the repository so that the demand reconstruction, the scenario Monte Carlo and the merit-order, missing-money and unit-commitment tests can be reproduced end to end from the raw inputs. The unit-commitment specification, its representative-day reducer and its validation gates are documented in the public repository (Data availability). A data-availability statement and a CRediT author-contributions statement accompany the submission; all derivations, per-country tables and the full input-parameter registers are provided in the public repository (Data availability).

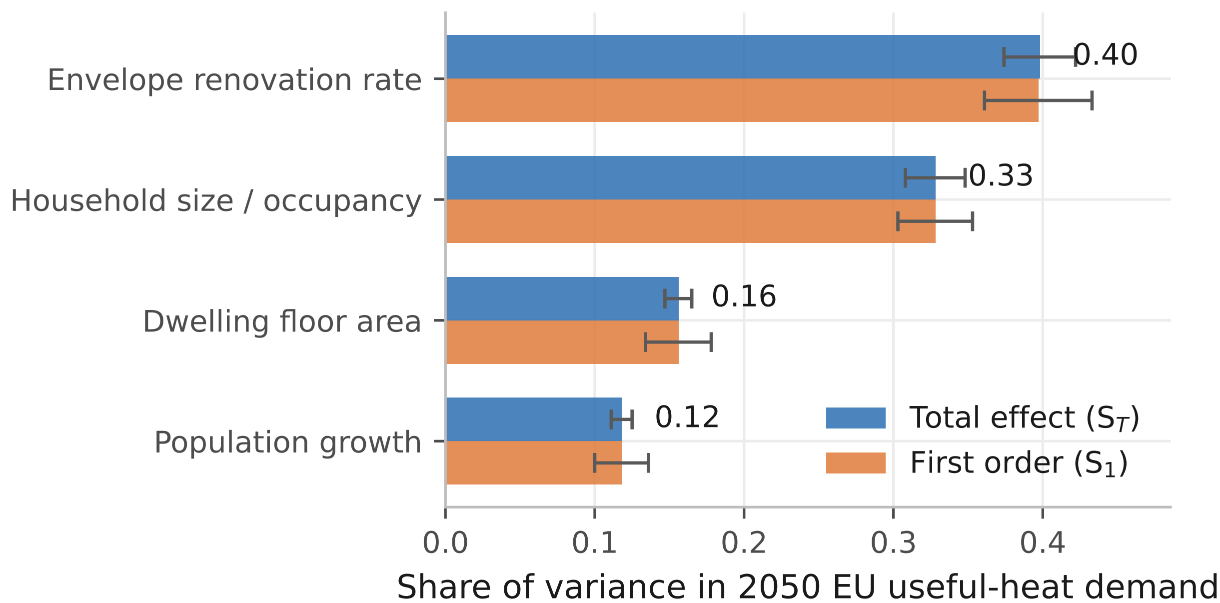

Figure S3: **Demand-driver Sobol sensitivity indices, 2050 EU useful heat.** Saltelli design, $N = 2{,}048$; drivers are the envelope-renovation rate, occupancy, dwelling size and population. Error bars are bootstrap 95 per cent confidence intervals over 100 resamples. The printed value on each pair is the total-order index S_T, which is near-equal to the first-order index for every driver here.

S2 Results and discussion

Cost and policy set the decarbonisation pathway for residential heat, and technology availability does not constrain it. The review literature treats the heat-pump-versus-hydrogen ranking as settled on cost. On the accounting used here, in which equipment capital is annualised over the heat a machine delivers, each carrier is charged its own last-mile network, and hydrogen is charged the seasonal store its winter-weighted load requires, that ranking holds across most of the study area without holding everywhere, and the exceptions concentrate where seasonal storage is cheap. What the results below add is therefore the spatial and system-level resolution of that contest (where, across 29 countries, and by how much each technology leads), an account of the storage geology and the fiscal asymmetry that together produce the exceptions, and a reframing of hydrogen's residual role from an energy question into a capacity one, tested across three dispatch arenas.

We establish the demand, technology-mix and emissions spine, then apply the same three tests the body applies, namely a building-level base-load cost comparison, a merit-order dispatch across three arenas, and an investment-feasibility test on the peaking niche. We close with the network-investment and least-cost cross-checks, the standalone Switzerland case, and the robustness of the whole.

The 200-sample Monte Carlo carries the demand, technology-mix and emissions spine across the four multi-lever scenarios (Current Policies, Stated Policies, Net Zero and H2 Push; Section S1), and those quantities are reported as medians with percentile bands. The base-load cost comparison, the dispatch tests, the investment-feasibility test and the least-cost program are evaluated at central parameters, and the parameters that decide each of them are swept deterministically over their plausible range, never sampled. Those sweeps are reported inline beside the central figures they qualify, and each result below states the basis it sits on.

S2.1 Heat demand, technology mix and emissions

Useful-heat demand across the 29-country study area is sticky (it barely falls despite renovation) because envelope renovation barely outpaces the growth of the building stock. In the 2025 base year we estimate it at 3,827 TWh per year; the underlying bottom-up reconstruction reproduces the Hotmaps 2015 regional total to within −0.8 per cent in aggregate (Section S1). Under Stated Policies, in which announced renovation is delivered at roughly today's EU pace, median demand declines only modestly, to 3,762 TWh by 2030, 3,626 TWh by 2040 and 3,483 TWh by 2050, a cumulative fall of 9 per cent; the 80 per cent Monte Carlo interval at 2050 spans 3,367 to 3,589 TWh.

Demand is held at the 1991 to 2020 climate normal throughout, which makes every trajectory above a frozen-climate one. Overlaying the projected decline in heating degree days lowers the 2050 Stated Policies level from 3,483 TWh to 3,338, 3,235 or 3,102 TWh on a low, RCP4.5 or RCP8.5 warming path, deepening the fall from 9 per cent to 13, 15 or 19. The overlay applies a study-area heating-degree-day decline of 0.20, 0.35 or 0.55 per cent a year to the space-heating share of useful heat alone, leaving hot water untouched, and it is applied to the finished trajectory rather than inside the projection, so it bounds the effect rather than resolving it. The rates bracket the 5 to 15 per cent mid-century decline the European degree-day literature projects across emissions pathways [40]; the middle rate is RCP4.5-consistent and the high one RCP8.5-consistent, whereas the low rate is our own conservative bound and not a published pathway. The overlay runs in one direction only, which is why the frozen-climate path is the conservative choice for a study whose results turn on winter peaks.

Because per-country envelope rates are sampled with a shared EU-policy correlation ($\rho = 0.5$; Section S1), this aggregate band is about 2.2 times wider than a naive independent-draw band, reflecting the limited cross-country diversification of a common policy environment. The four scenarios span a wide demand range, from a 5 per cent fall under Current Policies (3,651 TWh) to 31 per cent under Net Zero (2,660 TWh), with H2 Push intermediate at 3,154 TWh. Only the deep-renovation Net Zero pathway reaches the IEA Net Zero range of a 30 to 55 per cent reduction by 2050 [37]. Demand is therefore a supply-side question. The transition is decided by what meets the load; how much it shrinks matters far less.

The scenario lever moves the technology mix; cross-country variation in the stock barely enters. All four scenarios start from the same reconstructed 2025 mix on a useful-heat basis (heat pumps 8 per cent, gas 44 per cent, oil 14 per cent, biomass 17 per cent, district heat 11 per cent, resistance 6 per cent) and interpolate towards scenario-specific 2050 targets.

Three conventions govern those shares. The 8 per cent is a useful-heat share and is lower than unit-count figures on an equipment basis. Grouped shares are medians of the summed group across draws, taken with absent technologies counted as zero. A set of per-technology medians does not in general sum to one even when every individual draw does. At 2050 the sums run from 97.5 to 100.1 per cent; the widest gap on the whole path is Net Zero in 2040, whose raw sum of 93.5 needs a 6.5-point renormalisation. The reported shares are renormalised to 100 and we say so wherever they appear.

By 2050 the heat-pump share climbs from 29 per cent under Current Policies, through 37 per cent under Stated Policies, to 49 per cent under Net Zero, while residual fossil falls from 40 per cent to zero at the median under Net Zero (Fig. 3 of the main text); fossil heating survives in 49 of the 200 Net Zero draws, at a mean of 2.6 per cent across the run. District heat expands significantly in the deep pathway, to 31 per cent under Net Zero. The scenario lever moves the 29-country heat-pump share by 20 percentage points, from 29 to 49 per cent; within any one scenario, national shares vary by about 4 to 8 points between the tenth and ninetieth percentiles of the 29 countries.

Even in the deliberately hydrogen-favourable H2 Push world, hydrogen reaches only 8.5 per cent of useful heat at the realised median, against an 8 per cent scenario ceiling that itself sits below the 9.3 per cent the merit order would support, while heat pumps still lead at 39 per cent. Hydrogen's share is set by the scenario ceiling rather than by its levelised cost, and it does not move with the cost results reported below. Cheap hydrogen[2] grows the niche but does not change the architecture of the transition.

Where demand is sticky, emissions are not (Fig. S4, Table S5). All four pathways start from the same reconstructed 2025 baseline of approximately 644 $MtCO_2$, which is a property of the calibration that every scenario shares. The four fan out to 352, 231, 10 and 123 $MtCO_2$ by 2050 under Current Policies, Stated Policies, Net Zero and H2 Push, reductions of 45, 64, 98 and 81 per cent. The Net Zero band is not a smooth distribution and should not be read as one.

The announced replacement ban is triggered by a single scenario-wide draw crossing a fixed threshold, so the 200 draws split into two branches, 151 in which the ban binds, spanning 8.0 to 12.7 $MtCO_2$, and 49 in which it does not, spanning 55.0 to 136.9. No draw falls between 12.7 and 55.0, so 42 $MtCO_2$, two-thirds of the width of the reported $[9, 72]$ interval, carries no probability mass at all. The reported median of 10 is the median of the lower branch and the upper decile of 72 sits inside the upper branch. Net Zero delivers about 10 $MtCO_2$ with probability 0.76 and about 70 otherwise, both figures being branch medians, rather than a right-skewed distribution centred

[2] H2 Push runs the rapid hydrogen price path, whose 2050 delivered prices span about €22 to 39/MWh across the 29 markets, cheapest in Ireland on the green-electrolysis route. The central path, which the other three scenarios use, spans about €39 to 66/MWh against the €50/MWh north-west-Europe hub. This is the delivered figure inclusive of transport and exclusive of last-mile distribution, which the levelised cost charges separately (Section S1.3); lower values arise only at the production gate, before transport is priced in.

on 10. We retain the decile band for comparability with the other three scenarios and flag it here as a mixture.

The 10th-to-90th-percentile Monte Carlo ranges of adjacent scenarios do not overlap at 2050, though they do along the mid-path, so the 2050 ranking is robust to within-scenario parameter uncertainty; the four scenarios are distinct lever settings and none is a draw from a single population, so their separation is by construction.

The mechanism is decisive for the paper. The same technology set is available in every scenario, so *ambition separates the outcomes, and technology availability does not.* The 2050 gap between Current Policies and Net Zero, of roughly 342 $MtCO_2$ per year, is the emissions value of closing the implementation gap in residential heat, and it is a value that no amount of technology optionality delivers on its own. Because every scenario shares the same 2025 demand anchor, the ranking and the sign of this gap are invariant to that anchor; the absolute levels (the ≈644 $MtCO_2$ baseline and the ≈342 $MtCO_2$ gap alike) scale with it and inherit its uncertainty (Section S1), so we read the direction firmly and the magnitudes as anchored quantities.

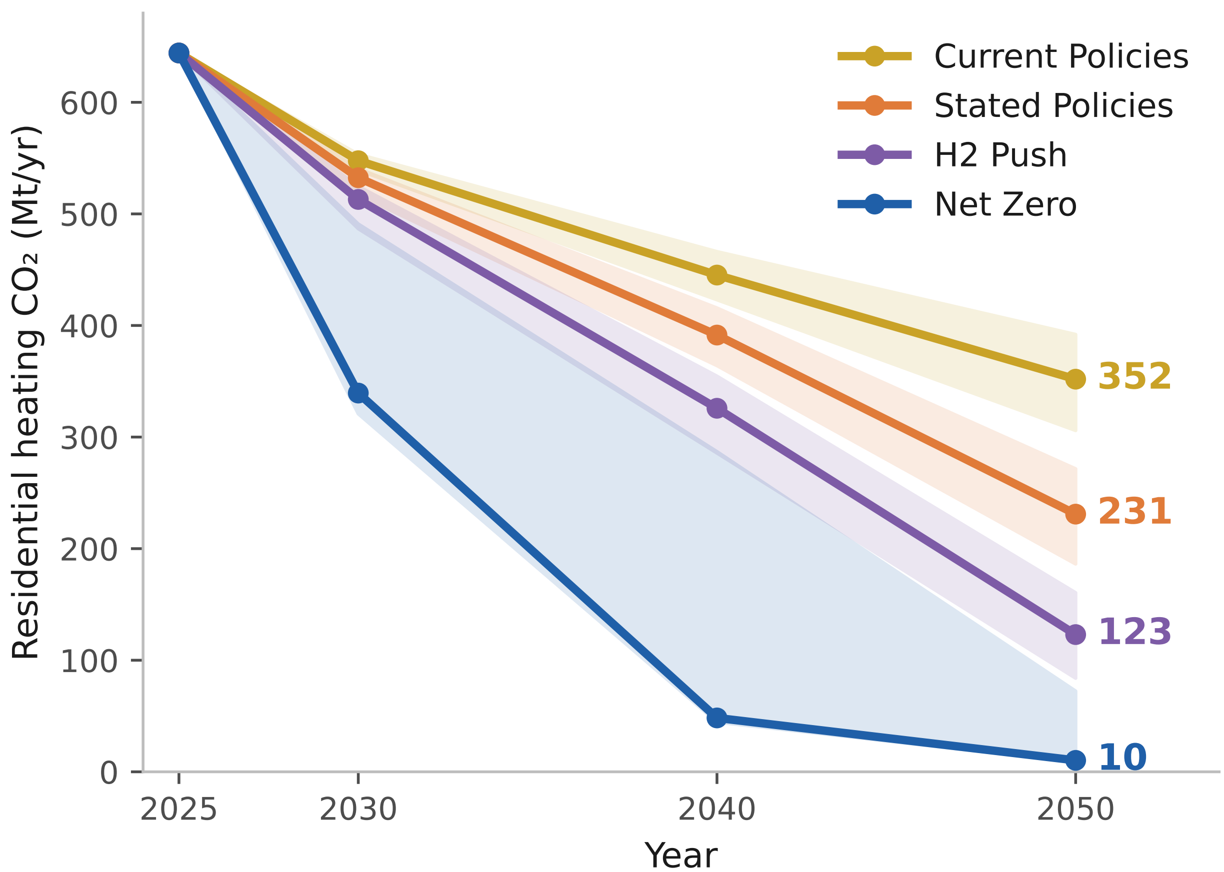

Figure S4: **Residential-heating CO_2 by scenario, 2025 to 2050.** Median path with the 10th to 90th percentile Monte Carlo band.

Note: all four start from the same reconstructed 2025 baseline (≈644 $MtCO_2$), and the bands of adjacent scenarios separate by 2050 while overlapping earlier. The Net Zero band is drawn solid like the other three and should not be read that way, because it is a two-branch mixture whose middle two-thirds carry no draws at all.

Table S5: **Buildings-sector CO_2 emissions by scenario and year.** In $MtCO_2$, with reductions against the reconstructed 2025 baseline of ≈644 $MtCO_2$, which all four scenarios share.

Scenario	2030		2040		2050		2050 range
	$MtCO_2$	Reduction	$MtCO_2$	Reduction	$MtCO_2$	Reduction	[10th to 90th]
Current Policies	548	−15%	445	−31%	352	−45%	[306 to 392]
Stated Policies	532	−17%	391	−39%	231	−64%	[186 to 272]
Net Zero	339	−47%	48	−93%	10	−98%	[9 to 72]
H2 Push	513	−20%	326	−49%	123	−81%	[84 to 160]

All four scenarios share the common reconstructed 2025 baseline (≈644 $MtCO_2$); every reduction is computed against it. The base year is a property of the calibration and carries no scenario lever, so the four are identical there by construction. Emission factors: IPCC AR6 (combustion), Ember 2024 (electricity actuals), EEA Fit-for-55 (projections).

Source: Authors' contribution.

S2.2 Base-load levelised cost of heat

The cost ordering among heating technologies is still open in 2050, and the convention used for the capital charge is what keeps it open. Air-source heat pumps and gas boilers sit at parity across the study area in 2025 (€155.8/MWh against €156.0/MWh at the median), with the heat pump ahead in 16 of the 29 countries. That reflects post-2022 residential gas prices (a Eurostat H1 2025 EU average of €114/MWh including taxes) set against taxed retail electricity and seasonal coefficients of performance (COP) of 2.8 to 4.1 across the existing air-source fleet; applying the model's climate correction the effective 2025 range across the 29 countries is 2.8 to 5.1, the upper end being ground-source units in the warmest markets.

Ground-source units are the cheaper heat pump in none of the 29 markets in 2025 and in 6 by 2050, so the air-source machine carries the comparison throughout. By 2050 the heat pump moves ahead of gas, at €118.5/MWh for the air-source unit and for the best available heat pump per country alike, against €132.7/MWh for gas at the median (Fig. 4 of the main text), with district heat cheaper than either at €96.5/MWh on a basis that figure does not carry. ETS2 widens the gas gap from 2027. Each €50/tCO_2 of carbon price adds about €11/MWh to the gas-boiler levelised cost of heat against almost nothing on the heat pump, because electricity carries only the increment above its 2025 carbon cost, floored at zero; that increment has a median of about €0.7/MWh across the 29 countries in 2030, reaching €5.1/MWh in Poland, and is zero at the median from 2040 and zero in every market by 2050 as the grid decarbonises. The central path reaches €122/tCO_2 by 2030 and €200 by 2050.

The heat-pump-versus-gas count therefore sits on the carbon path and moves with it, from 19 of the 29 markets on the low path through 24 on the central to 29 on the high, and every figure quoted here is the central path. Carbon pricing leaves the 2050 heat-pump-versus-hydrogen comparison unmoved, and the reason is dated rather than structural. Electricity carries the increment of its carbon cost above the 2025 level, worth a median of about €0.7/MWh of useful heat at 2030 on the central path, and that increment reaches zero in every market by 2050 as the grid decarbonises. Every hydrogen count reported below is therefore identical under the low, central and high carbon paths at 2050, while the 2030 and 2040 counts are not.

The hydrogen boiler follows the gas boiler down and, on the corrected capital and storage accounting, closes most of the distance to the heat pump without overtaking it. Its LCOH falls from €299.3/MWh in 2025 to €122.8/MWh by 2050 at the study median, above the €118.5/MWh best heat pump and below the €132.7/MWh gas boiler.

The base-load levelised-cost comparison asks, specifically, in how many of the 29 countries a hydrogen boiler undercuts the best heat pump in 2050. Under the central hydrogen trajectory, with each carrier charged the last-mile network its own delivery requires and hydrogen charged the seasonal store its winter-weighted load requires, the answer is 7. The heat pump leads in the remaining 22. Table S6 gives the full country ledger behind that count.

Table S6: **Base-load levelised cost of heat by country, 2050.** Best available heat pump, hydrogen boiler and the gap between them, in €/MWh of useful heat, sorted by the gap. A negative gap means hydrogen is the cheaper of the two clean options, which holds in 7 of the 29 markets.

Note: each median is taken down its own column, so the median gap is the median of the per-country gaps rather than the difference of the two medians, which here differ more than threefold. Gaps are rounded from full precision.

Country	Best heat pump	H_2 boiler	Gap (H_2 − HP)
DK	140.8	111.1	−29.7
DE	139.3	114.2	−25.1
UK	122.9	103.0	−20.0
NL	118.5	109.3	−9.1
BE	138.4	130.9	−7.5
IE	125.6	119.8	−5.8
PL	110.6	108.8	−1.8
CH	122.4	128.0	+5.6
FR	123.3	129.9	+6.7
AT	106.9	114.9	+8.0
LU	125.0	135.4	+10.4
FI	87.1	97.6	+10.5
CZ	108.3	122.1	+13.8
IT	137.5	151.5	+14.0
EE	80.8	96.3	+15.4
SE	85.2	102.5	+17.4
LT	80.0	99.5	+19.5
LV	77.1	98.7	+21.7
SK	99.0	125.3	+26.2
RO	85.0	111.9	+26.9
SI	107.4	137.2	+29.7
ES	126.5	158.3	+31.8
PT	129.9	166.6	+36.6
HR	98.5	138.4	+40.0
BG	81.9	122.8	+40.9
EL	130.2	177.7	+47.5
HU	65.1	126.5	+61.3
CY	196.9	284.5	+87.6
MT	187.6	286.2	+98.6
Median	118.5	122.8	+15.4

Source: Authors' contribution.

Denmark is the widest hydrogen win, at a hydrogen-minus-heat-pump gap of −€30/MWh, where cheap wind hydrogen and cavern storage meet costly, heavily taxed electricity, and it is followed by Germany (−€25), the United Kingdom (−€20), the Netherlands (−€9), Belgium (−€8), Ireland (−€6) and Poland (−€2). The heat pump leads by the widest margins in the warm south and east, at +€99 in Malta, +€88 in Cyprus, +€61 in Hungary, +€48 in Greece and +€41 in Bulgaria.

What decides the seven is the cost of the seasonal store, and that is what ties this result to the peaking arena below. Space heating draws about three-quarters of its annual energy between

October and March, a median winter share of 0.748 across the 29 countries with a range of 0.710 to 0.790 on the model's own monthly shape, while electrolytic production is far flatter. The mismatch has to be held from one season to the other, at an assumed €1.5/kg in salt-cavern markets and €3.0/kg elsewhere at about one cycle a year, and charging it on each country's own load-derived shift fraction adds about €12.1/MWh in Germany against €28.4/MWh in Spain (Section S1.3).

Five of the seven remaining hydrogen wins, including the four widest, are salt-cavern markets. The two that are not, Belgium and Ireland, survive on margins of €8 and €6/MWh while paying the higher rate. Hydrogen's holds concentrate in cavern markets, but geology does not select the set. Repriced at the non-cavern rate, only the Netherlands and Poland change hands. What decides the other five is the retail electricity tariff they are compared against. An accounting that charges no seasonal storage on the base-load test while charging it in full on the peak and dispatch tests prices the same molecule two ways, and the arenas then disagree. Charged consistently, they agree. The conclusions of the main text set out why that agreement corroborates less than it might.

Three summary statistics describe that distribution and we report all three, because they disagree. The unweighted median country gap is +€15.4/MWh, the unweighted mean is +€19.7/MWh, and weighting each country by its heat demand gives +€1.4/MWh. The distribution remains right-skewed (skewness +0.86), with Malta and Cyprus carrying the two largest heat-pump margins on 0.11 % of the study area's heat demand, which is why the unweighted mean sits above the unweighted median. Demand weighting moves the figure away from the median and towards parity, because the large, cold, high-electricity-price markets of the north-west are the ones where hydrogen leads. Of the three, the demand-weighted gap is the most decision-relevant, since it is the one that describes the heat actually supplied, and it is also the most favourable to hydrogen. The median remains the statistic comparable with every other level in this section, all of which are unweighted medians.

The implication for the base-load facet turns on what produces the remaining seven. Storage geology sets which markets are still contestable, and the fiscal treatment of the two carriers decides how the contest runs inside them. The gap is set by the ratio of two country-specific costs, and the two costs are not on an equal fiscal footing. By 2050 the median European household pays roughly €209/MWh for taxed retail electricity, against about €50/MWh for hydrogen priced delivered, which is untaxed for the reason set out in Section S1.3. That asymmetry is fiscal in origin, with no technological component, and it is what keeps the seven open once cheap storage has made them contestable. Where hydrogen wins, it wins against a levy, and equalising the treatment of the two carriers would move the comparison in the heat pump's favour, though the levy-equalised run that would size the effect is one we did not perform (Section S2.9). Whatever base-load cost case for hydrogen in buildings survives rests on storage geology and on the tax treatment of electricity, and the case against hydrogen for buildings runs through capital recovery in the peaking arena as well as through the levelised cost of base-load heat.

S2.3 Merit order across the three arenas

The levelised-cost comparison priced hydrogen as a base-load supplier. Its advocates instead price it as a *peaker*, the dispatchable source that serves only the cold-snap tip when the air-source COP collapses and the system is stressed. We reprice hydrogen on short-run marginal cost, the fuel-plus-carbon operating cost a system operator dispatches on, in three arenas: the building winter peak, the district-heat peak, and the power-sector peak. The scarcity electricity price against which the building peak is judged is derived from the power sector's own merit order; it averages €215/MWh under Current Policies, €240 under Stated Policies, €307 under Net Zero and €297 under H2 Push, because a high carbon price loads the marginal gas unit and makes the cold snap dearer in high-ambition worlds.

Hydrogen's win count rises with carbon-pricing ambition across all three arenas (Table S7, Panel A). On the symmetric basis, which charges hydrogen its own last-mile adder at the building peak and buys the district-heat operator's gas at the wholesale hub price, the counts are 0, 2, 7 and 14 of 29 countries at the building peak; 0, 0, 7 and 7 in district heating; and 0, 0, 7 and 7 in the power sector, always led by the cavern-endowed north-west. These are the counts the main text reports. On the asymmetric accounting, which charges hydrogen no last-mile network and prices the district-heat gas unit at the residential retail tariff, the first two arenas read 0, 4, 9 and 16 and 0, 5, 11 and 20; the power arena is untouched by either charge. Both bases are in the table, and the two paragraphs after next take each charge in turn. We report the power-sector count on the same short-run-marginal-cost criterion used throughout, namely the number of countries in which the hydrogen turbine's operating cost undercuts the carbon-priced gas peaker, evaluated on the peak-price series in which both turbines run at the same 40 per cent efficiency. A second, looser short-run cost series gives 0, 0, 7 and 15 on the same undercut test. That series carries the peaker profit and capital-recovery accounting, and it lets the two efficiencies diverge as each technology improves. The seven salt-cavern markets remain the decisive subset on the equal-efficiency series, which is the one the win counts use. On the looser series the count rises to 15 and the cavern confinement does not hold, so every statement below about cavern concentration is an equal-efficiency statement.

Two levers drive the ordering, and they act in sequence. Across Current Policies, Stated Policies and Net Zero it is the carbon price on the competing gas peaker, ETS2 at the building peak and ETS1 in district heat and power [41], that opens the niche. H2 Push then shares Net Zero's carbon path and adds only a faster decline in the delivered hydrogen price, so the further rise in the building and district-heat counts is the fuel lever; the carbon lever does not drive it. The power arena makes the split visible, holding at seven under both scenarios because the two turbines face the same carbon price there. The position of hydrogen in the stack is therefore set as much by the fortunes of its fossil rival as by its own cost.

The district-heat arena has a structural ceiling of 20, whichever basis it is priced on. Waste heat and other recovered sources sit at the bottom of the district-heat merit order at about €10/MWh, so a gas combined-heat-and-power plant is the marginal unit on a cold day in 20 of the 29 networks. In the remaining nine the hydrogen unit is itself the marginal source, so no gas unit is there for it to undercut and no accounting can return more than 20 on this test. On the residential-tariff basis hydrogen undercuts the gas unit in all 20; on the symmetric basis it does so in 7.

That difference is one price. The district-heat stack charges gas at the Eurostat residential retail tariff, taxes included, at a study-area median of €55/MWh in 2050, because that is the series the whole cost layer runs on. A district-heat operator buys wholesale. Repricing the gas unit at the €30/MWh hub price the power block already uses takes this arena from 0, 5, 11 and 20 across the four scenarios to 0, 0, 7 and 7. The count is a comparison of hydrogen against gas alone, so the electricity price the large heat pump pays does not enter it and the repricing is one-sided by construction. We headline the 7, because it is what a wholesale-buying operator would see and it is the number a district-heat investor should use; the 20 is reported alongside it as what the retail-tariff basis returns. The slice is also thin. The peak hydrogen could serve is the top 12 to 18 per cent of the load taken over the top 2,000 hours a year, which a peaking plant serves at about 518 equivalent full-load hours, so converting a full sweep of the 20 wins gives about 12 per cent of district heat weighted by demand, and about 3 per cent of all residential heat once district heat's 23.0 per cent share of 2050 useful heat under H2 Push is applied, 2050 being the year the sweep is run in.

The model prices the dispatchable tip of the 2050 power merit order and holds the low-carbon rungs at literature short-run costs. The figures in this paragraph are H2 Push; under Net Zero the same ordering holds with a narrower cavern advantage of €23 to 52/MWh-e and a non-cavern gap of €81/MWh-e. The fleet runs from near-zero variable renewables through nuclear (≈€12/MWh-e) and biomass (≈€55/MWh-e) to the dispatchable tip. There the position of the hydrogen turbine is *bimodal* across countries. At the cross-country median the hydrogen turbine (≈€286/MWh-e) sits about €30/MWh-e above the carbon-priced gas peaker (≈€256/MWh-e). In the typical market hydrogen does not undercut gas at all.

The seven power-sector wins are precisely the tail where it does. In the salt-cavern markets the hydrogen turbine falls to ≈€168 to 184/MWh-e, some €72 to 89/MWh-e below the gas peaker (Fig. 6 of the main text, which draws those seven and not the rest), while across the 22 non-cavern markets it sits about €36/MWh-e above. The dispatchable-tip ordering therefore flips sign with storage geology, and the direction of the headline claim (hydrogen undercuts the gas peaker only where salt caverns exist) is robust, while the median hydrogen turbine remains dearer than gas. We report the split, since a single representative margin would hide that the country distribution is strongly bimodal, so no one snapshot describes both the cavern winners and the non-winning majority.

The seven winning markets are precisely the salt-cavern countries, in descending advantage Denmark, the United Kingdom, the Netherlands, Romania, Poland, and France and Germany tied (Fig. 6 of the main text). The mechanism is the seasonal-storage adder σ in the short-run cost, approximately €50/MWh-heat where salt caverns exist and approximately €100/MWh-heat where they do not. The cavern geology follows the mapped potential [30]; the cost is an assumption of this study, a seasonal storage cost of €1.5/kg in cavern markets against €3.0 elsewhere at roughly one cycle a year. This ≈€50/MWh-heat geological difference is the largest single driver of the hydrogen unit's cost gap, with country-level variation in the delivered fuel price contributing the remainder of the cavern-versus-non-cavern difference. Because so much rests on a binary, Table S8 gives the classification and the charge it produces for every one of the 29 markets, so a reader who disputes either the split or the two rates can see precisely what turns on them.

The cavern concentration turns on two exogenous adders, the storage adder σ and the scarcity premium π_s, of which only the first can move the win count.

We therefore sweep the storage adder across nine values from €25 to 100/MWh-heat and check it against four scarcity premia from €30 to 120/MWh-e (Table S14). The count is identical at every premium, so the table carries one column and the invariance is stated rather than drawn as four copies of the same numbers. The count is unchanged by the scarcity premium, which adds equally to the gas and the hydrogen turbine and cancels from the comparison, and it holds at seven across storage adders of €50 to 75/MWh-heat, so the seven-market result is a robust feature of the storage geology. Read together, the base-load and peaking results give hydrogen the base-load cost comparison in 7 of the 29 markets, five of them cavern-endowed, and wins the power-arena peaking dispatch in the seven cavern-endowed markets alone. Both rest on the same seasonal-storage cost, which is why the two facets now agree, and neither gives hydrogen a self-financing route into building heat, for reasons the next subsection sets out.

S2.4 Capital recovery of the hydrogen peaker

Winning the dispatch is necessary but not sufficient. A private investor must also recover the peaker's capital from the few hours it runs. On a capital-recovery test, the merit-order peakers are not privately financeable from market rents, in any country or scenario, even where their operating cost wins the cold snap. Over the run window a winning market's peaker sees in 2050, and over

Table S7: **Operating-cost wins against capital recovery, by arena, accounting basis and scenario.** Counts of the 29 markets at 2050. Panel A is the merit-order dispatch on each of the two accounting bases and Panel B the capital-recovery test on market rent.

	Current	Stated	Net Zero	H2 Push
Panel A. Operating-cost wins (merit-order dispatch)				
Symmetric basis: hydrogen pays its €13.4/MWh median last-mile adder; gas enters district heat at the €30/MWh wholesale hub price. Headline.				
Building winter peak (vs gas peaker, cold-snap HP)	0	2	7	14
District-heat peak (vs gas combined-heat-and-power (CHP) plant, op. cost)	0	0	7	7
Asymmetric basis: hydrogen pays no last-mile network; gas is priced at the Eurostat residential retail tariff.				
Building winter peak	0	4	9	16
District-heat peak	0	5	11	20
Neither charge enters this arena, so one row serves both bases.				
Power peaker (vs gas peaker, operating cost)	0	0	7	7
Panel B. Capital recovery on market rent				
Recovers peaker capital from market rent	0	0	0	0

Turbine capital and short-run costs follow the Danish Energy Agency Technology Catalogue [14]. The power-peaker row counts countries where the hydrogen turbine's short-run cost undercuts the carbon-priced gas peaker on the equal-efficiency peak-price series. On the looser series used for the profit and capital-recovery accounting, which lets the two turbine efficiencies diverge, the Net Zero and H2 Push counts stand at 7 and 15 and are not confined to cavern geology.

Source: Authors' contribution.

Table S8: **The seasonal-storage classification the model applies, and the charge it produces, for all 29 markets.**

Country	Salt cavern	Rate (€/kg)	Full adder (€/MWh-heat)	Base-load adder (€/MWh-heat)
AT	no	3.0	100	24.3
BE	no	3.0	100	24.3
BG	no	3.0	100	24.3
CH	no	3.0	100	24.3
CY	no	3.0	100	28.4
CZ	no	3.0	100	24.3
DE	yes	1.5	50	12.1
DK	yes	1.5	50	12.1
EE	no	3.0	100	20.5
EL	no	3.0	100	28.4
ES	no	3.0	100	28.4
FI	no	3.0	100	20.5
FR	yes	1.5	50	12.1
HR	no	3.0	100	24.3
HU	no	3.0	100	24.3
IE	no	3.0	100	24.3
IT	no	3.0	100	24.3
LT	no	3.0	100	20.5
LU	no	3.0	100	24.3
LV	no	3.0	100	20.5
MT	no	3.0	100	28.4
NL	yes	1.5	50	12.1
PL	yes	1.5	50	12.1
PT	no	3.0	100	28.4
RO	yes	1.5	50	12.1
SE	no	3.0	100	20.5
SI	no	3.0	100	24.3
SK	no	3.0	100	24.3
UK	yes	1.5	50	12.1

Note: how the cavern binary is read and what the two rates assume are set out in Section S1.3. The base-load column charges each country the fraction of the full adder its own winter-weighted load requires, and divides by the year-specific boiler efficiency of 0.92 at 2050 where the two peaking arenas divide by 0.90 and charge the full adder, so it is not the full adder multiplied by the shift fraction.

Source: Authors' contribution.

about 139 full-load hours for the fleet as a whole, the inframarginal rent it earns (clearing price minus its own marginal cost) falls far short of its annualised capital requirement of about €51 to 66/kW-yr. At a clearing rent of €60/MWh, the scarcity premium, that rent covers only a small fraction of the requirement. The recovery fractions we quote are single-year ratios of rent to that year's capital charge, and they improve along the path as hydrogen cheapens and the carbon-priced gas rival grows dearer.

The study-area aggregate runs from deeply negative in the 2020s, when the hydrogen price is highest, through a first positive year in 2037, and then rises monotonically to 14.6 per cent in 2050 without turning over. It is still climbing at the horizon, so nothing here bounds recovery beyond 2050 and we make no claim about where it settles. Against that aggregate we quote about 9 per cent capacity-weighted across the winning markets in 2050, because a fleet figure credits a scarcity margin to countries where hydrogen loses the merit order and so overstates what a hydrogen peaker actually earns. The two are 2050 single-year statistics on different scopes, the aggregate the higher of the pair for that reason. We do not compute a winning-market trajectory over the whole path, so the comparison is stated at 2050 alone.

Behind the aggregate the markets differ widely, from about 5 to 40 per cent. The spread tracks utilisation. Every winner clears at the same €60/MWh scarcity margin, and run windows span 48 hours in Germany to 343 in Denmark. Recovery does not order on the same pair, because it also carries each country's cost of capital. The floor is Poland's 5.1 per cent on 56 hours at a 7.0 per cent weighted-average cost of capital, against Germany's 5.4 per cent on 48 hours at 4.0 per cent, and the ceiling is Denmark's 40.0 per cent (Table S11). Five of the seven winners build a 2050 peaker. The Netherlands and Romania win on operating cost but need no peaker under the three-hour standard, so their recovery is undefined.

This is the classic missing-money problem of low-utilisation peaking plant [29]. Cumulatively to 2050 the H2 Push hydrogen peaker loses about €357 bn against about €142 bn for an equivalent gas peaker, and crediting the same plant with a district-heat offtake barely helps[3] because a plant running only about 139 hours has little heat to sell.

The shortfall is not an artefact of the hydrogen-heavy scenario, and the scenario ledger runs the other way from the intuition. The cumulative gap widens as ambition eases, because the carbon-priced gas rival cheapens while the run window stays where the Stated Policies dispatch put it, so the hydrogen peaker loses the same hours at a worse margin. It runs from about €357 bn under H2 Push through €379 bn under Net Zero and €418 bn under Stated Policies to €563 bn under Current Policies, in every case about two-and-a-half to four-and-a-third times the loss an equivalent gas fleet takes over the same window. No setting in the sampled parameter space turns a hydrogen peaker into a self-financing asset.

The role this implies divides in two, and the division is the result. The firm-fleet accounting here is computed on the Stated Policies dispatch, the central announced-policy case, and then priced under each scenario's own fuel and carbon paths; the peaker's run window is therefore common across scenarios and only its margins move.

The firm dispatchable peaker fleet is about 278 GW across the EU-27, the United Kingdom and Switzerland, roughly 31 per cent of the 889 GW system peak. Two things qualify that ratio. The

[3]In the peaker economics the combined power-and-heat credit lifts the cumulative H2 Push position by about €8 to 16 bn (from −€357 bn to between −€350 and −€342 bn, under 5 per cent of the shortfall). Cumulative figures are undiscounted sums over the 25 years from 2025 to 2050, integrated across the milestone years on half-interval weights, so the terminal milestone carries a half-step. Extending the 2050 annual loss of about €13.3 bn/yr for a further five years, which makes the window 2025 to 2055, adds about €66 bn and gives about €424 bn. We report the 25-year figure throughout, because the model horizon ends in 2050 and the added years would sit at the most favourable annual rate in the run.

889 GW sums 29 national peaks that do not coincide, so it bounds the requirement from above in an interconnected system. And the 278 GW is an increment, sized on top of a firm combined-cycle gas fleet of about 195 GW that the dispatch holds fixed and runs ahead of the peaker, so total firm dispatchable capacity is about 473 GW, or 53 per cent of that peak; the peaking increment is the part hydrogen competes for. The energy share carries its own qualification, since the 4,513 TWh denominator grows non-heat demand by an assumed factor of 1.40 to 2050, which no model here produces. At a flat baseline the share would read about 1.2 per cent instead, and the capacity share is unaffected because the winter peak sets it. That requirement depends on how far the winter event is shared. Across cold-snap wind availabilities of 0.15, 0.25 and 0.40 the islanded requirement runs 291, 278 and 251 GW, against 267, 239 and 205 GW once the fleet is pooled across an EU copper plate, so the two-way grid spans 205 to 291 GW. A correlated continent-wide event sits nearer the islanded column, because pooling helps least when every country is short at once. That fleet is a large share of system peak capacity (Φ_{cap}, Eq. (S7)) and supplies well under 1 per cent of generated electricity (Φ_{en}), because the peaker runs only about 139 full-load hours a year on the reduced-form dispatch.

The explicit unit commitment puts a 280 GW fleet at 160 hours, because minimum stable output and ramp limits stop the cheaper combined-cycle plant following the residual load exactly and push the difference onto the fast peakers. Its fleet is the larger of the two by 2.8 GW because the commitment layer's own sizing pass builds a peaker in Romania, which the reduced form's duration test does not. That is the same mechanism that eases the missing-money position by about €1.3 bn against the reduced form, since at a common €60/MWh margin more running hours mean more rent. The easing belongs to the commitment constraints and not to the extra Romanian unit. The limit case already carries that unit and, with the constraints off, sits €1.1 bn *worse* than the reduced form, so switching the constraints on is worth €2.3 bn on its own, and it moves the capacity share not at all, because the unit commitment is anchored to the reduced-form fleet. We quote the reduced-form 139 throughout for consistency with the capacity accounting, and the unit-commitment 160 is the conservative direction for the recovery result. The capacity and energy shares of Eq. (S7) describe the same plant from its two sides.

Hydrogen's electricity role is accordingly a capacity role, and its energy contribution is negligible. It can serve as the clean firm backstop for the electrified heat load, and the markets where it wins hold most of the 278 GW firm peaker fleet, while it supplies a negligible share of generated kilowatt-hours [42]. The contribution of the spatial, system-level derivation is this capacity-not-energy distinction, which no single cost comparison delivers on its own.

The same molecule that wins operating-cost dispatch in as many as 20 of 29 markets in one arena (Table S7, Panel A) finances its own capital in none of them, in any arena or scenario (Panel B). Firm peaking capacity is sized to the fourth-highest residual-demand hour, a planning standard of three hours of lost load a year, so the 19 countries whose residual demand reaches that hour carry three hours by construction, nine carry none and Romania carries two. Those 19 build peaking capacity; the other 10, Romania included, build none, because a country whose residual exceeds the firm fleet in fewer hours than the standard allows to go unserved needs no peaker to meet it.

Adequacy is therefore an imposed standard, and what the model shows is that meeting it is unfinanceable from market rents. The difficulty is paying for the firm capacity, and the model's adequacy standard is met either way. A deliberate hydrogen-peaker build therefore implies a standing capacity payment, and the case for making that payment has to be argued on adequacy and emissions grounds. The base-load levelised-cost comparison does not settle it in either direction.

Two caveats bound the strength of this capacity verdict, and both would soften rather than reverse it. First, the 278 GW firm-capacity requirement and the €357 bn cumulative shortfall are computed on an *inflexible* electrified heat load. The model does not co-optimise demand-side flexibility, building thermal mass or storage sizing, so a flexible heat load able to ride through the cold-snap tip would shave the winter peak and shrink both the firm fleet and its capital bill.

We bound that reduction. Building thermal mass lets a heat pump pre-heat, moving part of its electricity draw out of the coldest hours into the hours either side, which is a shift, since the same heat is delivered. We therefore redistribute a share ϕ of the heat-pump load over a window of W hours, leaving the national baseline load, the weather path, the firm gas fleet, the storage state of charge and the three-hour planning standard exactly as the headline run leaves them. Only the shape of the heat-pump load moves.

A single smoothing operator would leave the answer open to the objection that it is a property of the operator, so we run three and report all of them. The *symmetric* kernel is a centred moving average, the most permissive of the three because load moves freely both backwards and forwards inside the window. The *pre-heat* kernel is one-sided; load may only be served earlier, never later, which is what thermal mass physically does, since a dwelling can be warmed in advance but cannot borrow heat from the future. The *capped* kernel is symmetric but clips the hourly displacement at a quarter of each country's mean heat-pump load, standing in for a finite store, and returns the clipped remainder to its original hour. All three conserve the annual total exactly.

Table S9: **Firm peaking requirement under a partially flexible heat-pump load.** Stated Policies, 2050, in GW against a 277.7 GW inflexible baseline, with the reduction in brackets.

Note: ϕ is the shiftable share of the heat-pump load and W the window, so $W = 13$ is up to six hours either side. The one-sided pre-heat kernel is the most effective and the finite store the least. The reduction is modest throughout because ϕ binds before the window does, and Table S10 shows it is nearly invariant to the assumed event length.

Kernel	ϕ	$W = 3$ h	$W = 7$ h	$W = 13$ h
Symmetric	0.10	277.1 (0.2%)	273.4 (1.5%)	269.3 (3.0%)
	0.20	276.2 (0.5%)	268.2 (3.4%)	261.6 (5.8%)
	0.30	275.0 (1.0%)	264.2 (4.8%)	253.0 (8.9%)
Pre-heat only	0.10	274.8 (1.0%)	268.5 (3.3%)	268.1 (3.4%)
	0.20	271.9 (2.1%)	260.4 (6.2%)	259.1 (6.7%)
	0.30	268.6 (3.2%)	248.4 (10.5%)	248.0 (10.7%)
Capped store	0.10	277.1 (0.2%)	273.4 (1.5%)	269.4 (3.0%)
	0.20	276.2 (0.5%)	269.7 (2.8%)	266.1 (4.2%)
	0.30	275.0 (1.0%)	268.0 (3.5%)	265.6 (4.3%)

Source: Authors' contribution.

At the most generous corner we sweep, 30 per cent of the heat-pump load shiftable across thirteen hours, which is well beyond what a housing stock mostly not built for it would deliver, the three kernels span 4.3 to 10.7 per cent, taking the fleet from 277.7 GW to between 265.6 and 248.0. We carry the largest of the three. The cumulative shortfall scales with the fleet, so the same fraction bounds it. What this does not do is substitute for a co-optimised flexible-load dispatch, which would size thermal storage against prices and would impose no shift window, and which remains future work. The 278 GW firm-capacity requirement and the €357 bn H2 Push cumulative shortfall are accordingly upper bounds that flexibility lowers by at most 11.6 per cent, the widest reduction over the event-length sweep.

One reading of that bound attributes it to the event: a multi-day lull, out of which no amount of intra-day pre-heating can move load. That reading restates the event length the dispatch assumes, and the sweep above varies the operator alone, so it cannot test it. Table S10 varies the event instead, taking the cold-and-still window from two days to ten while holding the seeds, the firm fleet, the Dunkelflaute depth and the sizing rule fixed, and recomputing the inflexible baseline at each length so the reduction is always measured against its own baseline.

Table S10: **Firm-capacity requirement against the assumed length of the synthetic cold-and-still event.** The flexibility reduction stays nearly flat across a fivefold change in event length, so the 278 GW headline carries about ± 9 per cent from this choice alone.

Event length (days)	Inflexible fleet (GW)	Flexible fleet (GW)	Reduction (%)
2	253.7	225.3	11.2
4	277.6	245.5	11.6
6	277.7	248.0	10.7
10	302.0	275.1	8.9

Source: Authors' contribution.

The sweep returns two results. The reduction is nearly flat, 8.9 to 11.6 per cent across a fivefold change in event length, so the bound is not an artefact of the six-day assumption. What limits it is the shiftable share ϕ, capped at 0.30, and the event length does not bind. Even a two-day snap, which a thirteen-hour window ought to blunt, yields only 11.2 per cent, because 70 per cent of the heat-pump load cannot move at all.

The sweep discloses a third thing. The *inflexible* fleet is itself a strong function of the assumed event length, running from 253.7 GW at two days through 277.7 at six to 302.0 at ten. The 278 GW headline therefore carries a band of roughly ± 9 per cent from this single modelling choice, wider than the flexibility correction it is usually qualified by. We report 278 GW on the six-day event throughout and note here what it would be on a shorter or longer one.

Second, the "no investor recovers capital" finding is partly definitional for any low-utilisation peaker. It is conditioned on the scarcity premium $\pi_s \in \{30, 60, 120\}$ €/MWh, a value of lost load of €8,000/MWh, about 139 full-load hours and the assumed absence of a capacity market. The absence of a capacity market is a modelling choice and describes no real market, so the counterfactual is worth setting out. Three of the seven cavern markets already run a mechanism. The United Kingdom has held technology-neutral capacity auctions since 2014, France has operated a decentralised certificate obligation since 2017, and Poland's centralised auction has run since 2018 [43]. Our energy-only baseline isolates what market rent alone delivers, and it is not a claim that these markets lack a mechanism. Where a mechanism does operate, the question this section asks becomes the design question of what the mechanism selects, and European law already imposes emissions conditions on that selection. Article 22(4) of the Electricity Regulation bars a capacity mechanism from paying a generator above 550 gCO_2/kWh, and from July 2025 bars capacity commissioned before July 2019 that exceeds both that rate and 350 $kgCO_2$ a year per installed kWe [44]. Both limbs are emissions conditions of exactly the kind selection would require. The annual limb does not reach the plant priced here at all, being scoped to capacity commissioned before July 2019 against a 2050 build, and on our own figures the rate limb does not bind either. On our emission factor of 0.202 tCO_2/MWh of gas, an unabated open-cycle turbine emits about 505 gCO_2/kWh-e at the 0.40 efficiency the win counts use, so it clears the rate limb because that rate sits above an open-cycle machine. It clears the annual limb by a wider margin still, for a reason that is this paper's own finding, since that test scales with running hours. At the fleet-average 139 the plant emits about 70 $kgCO_2$/kWe-yr and would have to run about 693 hours before the limb binds, and scarce running is what the plant is for. Were the annual limb extended to new capacity, selecting hydrogen would need it roughly a fifth of its present level. As drafted it reaches neither this plant nor any new one, so selection rests on the explicit premium quantified below.

Market rents alone cannot recover the capital of a ≈139-hour plant, so the policy-relevant question is what standing payment would. Net of the recovered rent, annualised capital leaves a gap that a payment of about €31 to 63/kW-yr (at the central scarcity premium) closes for the winning hydrogen peakers, whose market rent returns about 9 per cent of their €51 to 66/kW-yr annualised capital.

The payment is computed market by market, from each country's own recovery and cost of capital, and is lowest where the run window is longest (Table S11). Scaled at an unchanged scarcity margin, that ratio implies about 11 times the plant's 2050 run hours before it would finance itself unaided; because the margin would not persist over so many hours, the multiple bounds the gap without describing a reachable operating point.

Table S11: **The standing capacity payment a winning hydrogen peaker would need, 2050.** The seven salt-cavern markets in which the hydrogen turbine wins the winter peak on operating cost (H2 Push).

Country		Peaker	Run	Hydrogen		Gas	
	WACC (%)	fleet (GW)	window (h/yr)	Recovery (%)	Payment (€/kW-yr)	Recovery (%)	Payment (€/kW-yr)
Denmark	3.5	7.9	343	40.0	30.8	−34.8	52.0
United Kingdom	4.5	49.4	112	12.2	48.7	−10.4	45.9
France	5.0	37.2	71	7.4	53.3	−4.2	45.0
Germany	4.0	87.7	48	5.4	50.5	−3.0	41.3
Poland	7.0	22.0	56	5.1	63.1	−3.1	51.4
Netherlands	4.0	0.0	0	n/a	n/a	n/a	n/a
Romania	7.5	0.0	0	n/a	n/a	n/a	n/a
Capacity-weighted	n/a	204.1	n/a	8.7	51.2	−6.3	44.6

Note: recovery is the single-year ratio of inframarginal rent to that year's annualised capital charge, and the payment is the shortfall the rent leaves. The gas column prices the same plant at €450/kW against €600/kW for a hydrogen-ready turbine, and its recovery is negative because hydrogen sets the clearing price here. The Netherlands and Romania build no peaker, so their recovery and payment are undefined. Cells round independently, so the capacity rows sum to 204.2 against a printed 204.1.

Source: Authors' contribution.

The payment attaches to low-utilisation firm capacity of any fuel, and the comparison with gas has two sides worth separating. A gas peaker recovers a smaller share of its capital than hydrogen in the cavern markets, and its rent there is negative, yet it needs a slightly smaller payment in absolute terms because a gas turbine costs €450/kW against 600 for a hydrogen-ready one. Capacity-weighted the requirement is about €45/kW-yr for gas and 51 for hydrogen, and gas is cheaper in four of the five markets that build. A technology-neutral capacity payment in these markets would therefore procure unabated gas ahead of hydrogen, at the €600/kW we assume for a hydrogen-ready turbine; selecting hydrogen requires an emissions condition, whose implied cost is the payment difference divided by the gas generation it displaces, about €172/tCO_2 on top of the carbon price already in the gas plant's marginal cost.

Two things bound this figure, and they run in opposite directions. The first is the capex gap, which is what the whole corollary rests on. Swept from €450 to 700/kW against the gas turbine's 450, the hydrogen turbine drives the conditioning premium from −€192 to +€415/tCO_2 and changes sign between about €520 and 540/kW, a premium of roughly 18 per cent over gas. Below that break-even the corollary inverts. Hydrogen is the cheaper firm option on a per-kW basis and a technology-neutral payment procures hydrogen. Published premia for hydrogen-ready simple-cycle machines straddle this break-even, so the direction of the result is a property of the 33 per cent capital premium we assume, €600/kW against €450; the markets do not produce it.

The second is that we report this as a lower bound only. The gas peaker is priced inside a market whose clearing price hydrogen sets, so its rent comes out negative, between −€25 and −39/MWh. No operator would hold that position; it would withhold capacity instead. Flooring that rent at zero lowers the gas plant's requirement to about €42/kW-yr, widens the capacity-weighted gap and lifts the implied conditioning premium to roughly €239/tCO_2. The direction is unaffected and the reported figure is the conservative end of the range. The "no investor recovers capital" finding is a statement about market rents alone. These figures follow by arithmetic from the reported peaker economics and are tabulated per market in the public repository.

S2.5 Network investment by pathway

The levelised-cost layer charges each carrier its last-mile network per MWh, which keeps the technology comparison symmetric but hides the absolute capital at stake. We therefore cost the wires and pipes each pathway implies directly, in billions of euros cumulative to 2050 across the 29 markets, from the model's own 2025 to 2050 technology shares.

Three networks are costed. Electricity distribution reinforcement follows the added coincident heat-pump electric peak, which is the increase in heat-pump share between 2025 and 2050 multiplied by the country's heat peak, divided by the cold-snap coefficient of performance and scaled by a coincidence factor of 0.5, charged at €400, €800 and €1,600/kW-peak on the low, central and high bands. District-heat expansion follows the increase in district-heat share multiplied by dwellings, at €3,000, €5,000 and €8,000 per connection, the central figure being the same one Section S2.2 charges district heat when the connection asymmetry is closed. The hydrogen network follows the 2050 hydrogen-boiler share multiplied by dwellings, at a per-connection cost that blends the conversion and new-build bands on each country's gas-grid coverage, so a country with dense existing gas mains converts cheaply and one without it pays to build.

Table S12: **Cumulative network investment, 2025 to 2050, across the 29 markets.** In bn €, by pathway and by network. The central column uses the mid cost band for each network; the final column carries the low and high bands.

Note: on the central band the pathways span €249 to 506 bn. Cells round independently, so a row's three networks can sum to one short of its own total. The €74 bn hydrogen network in H2 Push sits alongside, and does not displace, €89 bn of electricity reinforcement in the same pathway.

Pathway	Electricity	District heat	H_2 network	Total (central)	Total (low to high)
Current Policies	62	183	4	249	143 to 423
Stated Policies	83	194	48	325	187 to 558
Net Zero	119	356	31	506	292 to 861
H2 Push	89	249	74	412	240 to 705

Source: Authors' contribution.

Table S12 gives the result. On the central bands the pathways span €249 bn under Current Policies to €506 bn under Net Zero, and the band width within any one pathway is wider than the spread between pathways, so the ordering carries further than the levels. District heat dominates every pathway, because a connection is charged per dwelling while reinforcement is charged per kilowatt of coincident peak. The figure that bears on this paper's question sits in the H2 Push row, where €74 bn of hydrogen network stands alongside €89 bn of electricity reinforcement and does not displace it, since even in the pathway most favourable to the molecule the heat-pump share still rises. A hydrogen pathway for buildings adds a third network to be paid for.

Two limits apply. Dwelling counts come from the bottom-up stock, whose levels run some 20 to 30 % high in a few countries; that is a level bias which cancels in the cross-pathway comparison and is in any case dominated by the roughly twofold cost-band width. And the bill is a deepening cost only. Transmission, storage and generation capital sit outside it, so these are not whole-system pathway costs and should not be read as such.

S2.6 Switzerland as a standalone case

Switzerland is the standalone case, and both facets settle the same way there. On the cost model, the Swiss heat-pump LCOH runs at €160.5/MWh (air-source) and €175.8/MWh (ground-source) in 2025 and falls to a 2050 band of €122.4 to 131.1/MWh, against a 2050 Swiss hydrogen-boiler LCOH of €128.0/MWh; the Swiss gas boiler, for reference, starts at €216.0/MWh in 2025 and reaches €165.9/MWh in 2050 (Fig. 7 of the main text). The best Swiss heat pump is therefore €5.6/MWh cheaper than the hydrogen boiler on 2050 base load, which places Switzerland outside the seven hydrogen wins. The base-load case does not reverse. Switzerland has no salt caverns, so it pays the €3.0/kg seasonal storage rate on any hydrogen burned for heat, and that rate is what restores the heat-pump ordering against expensive, heavily levied Swiss retail electricity. Charging no seasonal store on the base-load test reverses the Swiss ordering, so the storage charge is the term that decides Switzerland.

On the peaking side, Switzerland has no salt caverns either, so the seasonal-storage adder is €100/MWh-heat. The endogenous coupling confirms that the hydrogen peaker loses the Swiss winter peak in all four scenarios, at €134.8 to 235.9/MWh-heat against a heat-pump peak of €62.3 to 90.3/MWh-heat, priced out by storage cost alone.

The strategic answer therefore rests on one physical fact in both tests. Switzerland has no geological route to cheap seasonal storage, so a domestic hydrogen firm-capacity role is closed to it, and its access to European supply is best directed at hard-to-abate industry, heavy transport and winter power adequacy, where the molecule substitutes for costly or absent alternatives. The Swiss base-load margin is narrow enough that the tax treatment of retail electricity still bears on it, so we do not read the €5.6/MWh as a pure technology result either.

S2.7 Significance, robustness and demand-driver sensitivity

The base-load result is conditional and we report it as such. The green-hydrogen price path sets the win count directly. On the rapid path the heat pump is cheapest in 13 of the 29 markets, on the central path in 22, on the slow path in 26 and on the stranded path in all 29. Because each scenario carries one of those paths, the base-load ordering is a property of the hydrogen price assumption rather than of the policy scenario, and the carbon price does not enter the comparison at all.

The count is also a boundary statistic, and we state how close that boundary runs. Seven of the 29 countries sit within €10/MWh of it: Poland at −1.8, Switzerland at +5.6, Ireland at −5.8, France at +6.7, Belgium at −7.5, Austria at +8.02 and the Netherlands at −9.1, with Luxembourg at +10.4 and Finland at +10.5 the nearest beyond. A uniform €8/MWh shift in either direction, which is smaller than several of the sensitivities reported in this section, would move the count between 20 and 25; a €10/MWh shift moves it between 19 and 26. Only Poland lies within €5/MWh, so the seven hydrogen holds are not uniformly marginal, but the count should be read as a statistic with about five markets of play in it and not as a sharp threshold. The margins themselves are the more robust quantity, and we report them per country alongside the count.

The seasonal-storage charge sets the count just as directly, and because the headline is close to linear in it we sweep it continuously, not at three points. Writing the charged fraction as a multiple of the load-derived shift fraction (median 0.248, country range 0.210 to 0.290), the count of markets in which the best heat pump is cheapest and the median country gap run as follows.

Multiple of load-derived fraction	Charged fraction	Heat-pump wins	Median gap (€/MWh)
0 (nothing charged)	0.000	12	−5.1
0.25	0.062	15	+0.0
0.50	0.124	19	+5.2
0.75	0.186	21	+10.3
1.00 (adopted)	0.248	22	+15.4
1.50	0.372	25	+26.1
2.00	0.496	26	+38.0
4.03 (full seasonal adder)	0.999	29	+81.7

The count is monotone and close to linear in the charged fraction over the first half of the range. It falls to 19 at half the adopted fraction, to 15 at a quarter, and to 12 only when the store is not charged at all, so the summary of the base-load verdict is 22 markets on the adopted basis and 15 to 26 across the fractions we consider plausible. Halving the fraction is defensible if electrolytic supply carries any seasonality of its own, or if summer molecules come off a shared backbone rather than a dedicated store, and readers who prefer that reading should take 19 rather than 22. What no plausible fraction reaches is either extreme, the 28-of-29 heat-pump sweep that the uncorrected capital accounting produces or the 17-of-29 hydrogen majority that the zero-storage arm produces.

The hydrogen network treatment shifts the count almost as sharply. Sweeping the delivery build from free reuse of the existing gas grid, through retrofit conversion and the coverage-weighted blend used as the baseline here, to new-build dedicated pipe, with every arm carrying the seasonal store so that all of them are comparable both with one another and with the headline, the EU-mean hydrogen-minus-heat-pump gap runs from €0.0 through +€8.0 and +€19.7 to +€26.9/MWh, reaching +€43.1/MWh at the new-build high bound. The count of markets at or below parity for hydrogen falls from 12 to 9, 7, 3 and finally 2. Only Denmark and Germany survive the most expensive treatment. Denmark runs from −€44.3/MWh under free reuse to −€28.0/MWh under new build and −€18.2/MWh at the high bound, never crossing zero; Germany runs from −€33.2 to −€16.2 and −€6.0/MWh.

The network treatment therefore does more than shrink hydrogen's lead. It closes that lead in five of the seven markets hydrogen holds under the default. Hydrogen's base-load case outside Denmark and Germany depends on inheriting distribution assets it does not pay to build.

The mirror test on the electricity side charges the heat pump the high end of the low- and medium-voltage reinforcement band, €4.4 to €38.2/MWh against the €2.2 to €19.1/MWh central value used throughout. Doing that while charging hydrogen no network at all, which is the most hydrogen-favourable configuration the model admits, reverses the headline. Heat pumps then stay ahead of hydrogen in only 10 of the 29 markets, Bulgaria, Estonia, Croatia, Hungary, Lithuania, Latvia, Romania, Sweden, Slovenia and Slovakia, though ahead of gas in 21, at a mean headroom of −€6.2/MWh (median −€3.1). That arm is deliberately incoherent, since it charges one carrier for its network and exempts the other. Restoring hydrogen's own network charge, so that both carriers pay, returns the count to 22 markets at a mean of +€13.6/MWh and a median of +€12.3/MWh. The headline count is therefore reproduced at the top of the reinforcement band and not only at its centre, which makes the base-load ordering robust to how expensive electricity distribution proves to be and fragile to whether hydrogen is charged for distribution at all.

What is robust is the peaking verdict, and the three modelling extensions leave it intact. The explicit unit-commitment dispatch agrees with the reduced-form merit order to within 0.33 per cent on dispatched energy and 0.40 per cent on peak capacity in each of 28 countries. The twenty-ninth, Romania, is a structural disagreement. The reduced form applies the three-hour standard to a residual duration curve and finds no peaker needed. The unit commitment must also respect minimum stable output, ramp limits and the reserve margin, so it builds 2.8 GW and eases the missing-money loss by about €1.3 bn. The myopic rolling-horizon foresight leaves the cost-optimal mix and the retrofit arm's 2050 zero unchanged, and the endogenous Switzerland coupling reproduces the standalone Swiss peaking verdict.

The heating-to-power feedback (derived scarcity price, heat-pump competitiveness and electrified load) is closed as a fixed point only for Switzerland. Adding Swiss heat-pump load alone barely moves the winter price, from €240 to 246/MWh, but crediting the Swiss reservoir fleet as firm winter capacity in the full endogenous run lowers it to about €141/MWh, so the coupling is not negligible in the direction that matters for heat-pump competitiveness; the EU-wide branch prices the heat-pump peak against fixed scenario grid paths, so the residual circularity at EU scale is bounded by the Swiss case, not solved at scale.

The demand side carries its own variance-based screen. A Sobol decomposition (Saltelli design, $N = 2{,}048$) of 2050 EU useful-heat demand attributes its variance to the four demand drivers: the envelope-renovation rate ($S_T = 0.40$), occupancy (0.33), dwelling size (0.16) and population (0.12), with first-order indices approximately equal to the total-order indices, indicating a near-additive response over the sampled ranges (Fig. S3). Those ranges are deliberately unequal, each set to its own driver's plausible span: $\pm 35\,\%$ on the renovation rate, $\pm 5\,\%$ on occupancy, -2 to $+5\,\%$ on dwelling size and $\pm 3\,\%$ on population. The ranking therefore reflects both the model's response and those widths, and the renovation rate leads partly because its band is the widest; on equal relative bands the ordering changes.

The result to take from the screen is that the renovation rate contributes the most variance *over its own plausible range*, which is why the demand reconstruction treats it and occupancy as first-class drivers, and not that it is intrinsically the most influential parameter. This is a decomposition of demand-side variance only. We are candid that it does not extend to the axes that carry the hydrogen and emissions headlines. The carbon price, the green-hydrogen trajectory, the WACC, the ETS2 pass-through and the fuel-price draws enter the headline results through the levelised-cost and merit-order layers, as several correlated economic levers that a Monte Carlo percentile band and a one-at-a-time tornado alone cannot disentangle.

We therefore run the global cost-side screen directly. Complementing the demand-side Sobol decomposition, we compute standardised rank-regression coefficients (SRRC) across all $\sim$17 sampled parameters (the fossil, heat-pump, district-heat and hydrogen 2050 target shares, the demand-reduction and demand-shape levers, the technology-uptake exponents, the ground-source share, the ETS2 pass-through, the discount rate and WACC, the heat-pump-feasibility multipliers, and the per-carrier gas, electricity and hydrogen fuel-price multipliers) on the real NUTS3 Monte Carlo ($N = 192$ draws, H2 Push scenario, output the 2050 buildings-sector CO_2; SRRC is valid on this random-draw design). The screen is decisive (Table S13). The fossil-phaseout ambition lever dominates the 2050-emissions variance (SRRC = 0.98; rank-R^2 = 0.95), while every economic-price parameter is negligible. The delivered hydrogen-price multiplier carries $|\mathrm{SRRC}| = 0.009$, the electricity price 0.015, the gas price -0.007, the ETS2 pass-through 0.011 and the discount rate 0.002, all with $|\mathrm{SRRC}| < 0.02$.

We read that result narrowly, for reasons set out in the Methods. The price coefficients are near zero because the 2050 emitting share is price-inert by construction. That is a property of the model, and not evidence that prices are immaterial. Holding everything else fixed and scaling every fuel price by 0.6 or 1.6, or moving the discount rate across the full 3 to 8 per cent country range, leaves the 2050 gas-plus-oil share unchanged to four decimal places. The carbon price is not sampled in the Monte Carlo at all, so this screen carries no information about the carbon axis, and the sampled hydrogen and discount bands are much narrower than the scenario spans they resemble.

The screen also does not transfer across scenarios. Repeating it identically returns rank-R^2 of 0.930 and 0.942 under Current Policies and Stated Policies but 0.432 under Net Zero, where the fossil coefficient falls to 0.583 because the response there is a step, not a monotone function, and a rank regression is not a valid summary. What the screen supports is that within the sampled economic ranges the 2050 emissions ordering is set by policy ambition and not by the assumed price trajectories. It does not establish robustness over the full cost, carbon, hydrogen-price and discount space.

Table S13: **Global variance-based sensitivity of 2050 buildings-sector CO_2 emissions.** Every parameter the Monte Carlo samples, as standardised rank-regression coefficients (SRRC) over the NUTS3 Monte Carlo ($N = 192$ draws, H2 Push scenario; rank-$R^2 = 0.95$). One policy lever governs the 2050 emissions headline and every other axis carries $|\text{SRRC}| < 0.02$.

Sampled parameter	SRRC
Fossil-phaseout ambition (2050 fossil target share)	+0.977
Heat-pump feasibility, multi-family†	−0.019
Electricity-price multiplier	+0.015
2050 demand reduction†	+0.014
District-heat uptake exponent	−0.012
ETS2 pass-through	+0.011
District-heat 2050 target share	−0.010
Heat-pump 2050 target share	+0.010
Heat-pump uptake exponent	+0.010
Demand-reduction path exponent†	+0.009
Hydrogen-price multiplier	+0.009
Heat-pump feasibility, single-family	−0.008
Gas-price multiplier	−0.007
Hydrogen uptake exponent	+0.006
Ground-source share of heat pumps	+0.004
Hydrogen 2050 target share	−0.004
Discount rate / WACC	+0.002

†Sampled but inert in the production configuration, so the coefficient is noise on a variable the model does not read. Listed for completeness of the sampled set, and its rank is not influence.

Source: Authors' contribution.

A second robustness question concerns the seven-market power-arena result, which rests on two exogenous, single-source stylised values: the seasonal storage adder and the scarcity premium. We sweep the first over the real 2050 power merit order and report it against the second (Table S14). The count is invariant to the scarcity premium across its full €30 to 120/MWh-e band, and that invariance is algebraic, not empirical. The premium adds equally to the gas and the hydrogen turbine and so cancels exactly from the comparison that decides the win. It sets the winners' capital recovery and not the win set, which is why we report it as an identity and rest no robustness claim on it.

Along the storage axis the winning set is exactly the salt-cavern markets and the count holds at seven across storage adders of €50 to 75/MWh-heat, a 50 per cent increase on the central value. Outside that window the result does not hold. An eighth market enters at €45/MWh-heat, all 29 win at €38 and below, only two survive at €88 and none at €100. The sweep also scales the cavern and non-cavern adders together, so it tests the level of the storage penalty and holds the cavern-to-non-cavern ratio fixed at 2.0; the geological spread that actually drives the concentration is not varied. That ratio is the one parameter of the seven-market result this sweep cannot test, so its only support is external. A European techno-economic assessment of the same stores puts levelised salt-cavern storage at about \$0.8/kg against \$1.5/kg for porous media [45], a ratio near 1.9 against the 2.0 assumed here, which is the corroboration the conclusions cite. The seven-market result is therefore robust to the scarcity premium entirely, and it is a feature of the central half of the swept storage range only.

Table S14: **2050 power-arena hydrogen-win count against the storage adder and the scarcity premium, H2 Push.**

Note: the count is identical across the €30 to 120/MWh-e premium band, so one column is shown. The storage adder is the cavern-market value on the heat side, with its scale relative to the central value in parentheses. The winning set equals the seven salt-cavern markets across adders of €50 to 75/MWh-heat.

Storage adder (cavern, €/MWh-heat)	Power-arena H_2 wins	Winning set
25 (0.50×)	29	all markets
38 (0.75×)	29	all markets
45 (0.90×)	8	the salt-cavern markets plus one
50 (1.00×, central)	7	the seven salt-cavern markets
55 (1.10×)	7	the seven salt-cavern markets
62 (1.25×)	7	the seven salt-cavern markets
75 (1.50×)	7	the seven salt-cavern markets
88 (1.75×)	2	two salt-cavern markets
100 (2.00×)	0	none

Every count is invariant to the scarcity premium over €30 to 120/MWh-e; that premium governs the winners' capital recovery (Section S2.4), not which markets win.

Source: Authors' contribution.

What this analysis establishes divides into a robust part and a conditional part, and we keep the division explicit. The base-load count turns on three accounting choices, and no physical argument settles any of them. Sizing equipment capital to peak heat demand and charging each carrier the last-mile network its delivery requires, while charging no seasonal storage on the base-load test and charging it in full on the peak and dispatch tests, leaves 12 markets to the heat pump and opens a north-western exception of 17 markets to hydrogen. Pricing the same molecule the same way in every test, which is the basis adopted here, gives 22 for the heat pump and 7 for hydrogen. The count is a property of the accounting as much as of the cost data, which is why we state the accounting in full rather than report an interval around it.

What survives is the emissions result and the capacity result. The global cost-side screen closes the emissions headline and confirms that 2050 buildings-sector CO_2 is governed by policy ambition and not by any economic-price axis (Table S13). The capacity verdict survives untouched, since no hydrogen peaker recovers its capital from market rent in any country or scenario, and that finding rests on a comparison between two turbines at wholesale fuel prices, not on the levelised cost of base-load heat.

The policy reading that now rests on robust ground is accordingly narrower. Hydrogen in buildings is a support-contingent capacity option, and what remains of its base-load cost advantage reads as a joint consequence of cheap cavern storage and of taxing electricity while leaving the molecule untaxed. The levelised-cost ordering moves with the hydrogen price path, with the seasonal-storage charge and with the network-cost treatment.

S2.8 Least-cost program and the bounds on reading it

The four scenarios impose technology shares, so a parallel least-cost linear program minimising supply cost against a fixed demand path and a scope-1 emissions cap provides a cost-optimal cross-check. It is silent on hydrogen, for the reason given below. At the -90 per cent cap the 2050 least-cost mix is 52.9 per cent district heat, 45.4 per cent heat pumps (44.4 air-source and 1.05 ground-source), 1.4 per cent residual gas and 0.34 per cent hydrogen. On 2050 levelised cost alone the cheapest single technology is district heat in 25 countries, the hydrogen boiler in 2 and a heat pump in 2, on a three-way comparison that includes district heat, which the headline pairwise count excludes.

Three bounds should be read alongside that mix. District heat's lead rests on the unpriced connection charge, so its share is an upper bound; we do not re-solve the program with the charge applied and so cannot report how far the 52.9 per cent would fall, only that charging it moves district heat from cheaper than the best heat pump in 26 of 29 countries to 7. The program carries a per-country hydrogen ceiling, separate from the exogenous adoption setting of the four-scenario framework, and it selects exactly that ceiling in all 29 markets, so the 0.34 per cent aggregate is a ceiling outcome and not a cost verdict. The ceiling is 3 per cent in the Netherlands, 2 in the United Kingdom and zero in the other 27, which makes it an exclusion in most of the study area rather than a low cap. The program's emissions cap applies to each country against its own 2025 baseline with no cross-border pooling, whereas ETS2 is a pooled price, so where the cap binds it tests whether a national target is reachable and leaves open what a single EU-wide carbon price would deliver. The cap binds in Cyprus, Croatia, Poland and the United Kingdom at 2050, and in three more, Spain, Hungary and Romania, at some earlier point on the path, for seven in total. The agreement between the two frameworks is accordingly about direction alone.

S2.9 Limitations and the direction of each

These results are one decision input. The model compares technologies pairwise instead of co-optimising demand and supply, does not endogenise storage sizing or demand-side flexibility, and resolves each region to NUTS3 with three building types, averaging over the dense historic urban cores where any residual hydrogen niche would sit. Seven specific limitations bound the reported numbers, and we give each with the direction it pushes the result.

The weather. The first and largest bounds the whole power-sector half of this paper. Every figure in it, the 278 GW firm fleet, the 139 full-load hours, the €357 bn cumulative shortfall and the zero row of the capital-recovery test, is computed on one realisation of a single synthetic year, in which onshore wind is an AR(1) process forced to a quarter of normal over six days in mid-January in all 29 countries simultaneously. That event is perfectly correlated across the study area because we construct it that way, and its length is a choice. Two days gives a 254 GW fleet and ten days gives 302 GW against the 278 GW we report. An adequacy assessment would run thirty or more recorded weather years and report the requirement as a distribution, and it would derive the correlation from the data instead of imposing it. We do neither, and we benchmark none of these figures against a published European adequacy study. They should be read as the output of a screening model on a stated synthetic year, and a reader who wants an adequacy number should not take one from here.

The fiscal asymmetry. The second bears on the base-load comparison above all others. That comparison sets a household's taxed retail electricity price against hydrogen valued delivered and untaxed, so the ordering in the seven north-western markets where hydrogen still leads records the incidence of levies, network charges and taxes as much as it records any property of the two technologies. We hold the fiscal treatment of both carriers fixed at today's, which is a strong assumption over a 25-year horizon and one that policy could deliberately overturn in either direction. The levy-equalised comparison that would size the effect is not run here, so the seven surviving hydrogen leads should be read as conditional on the present asymmetric treatment.

Exogenous hydrogen prices. The third cuts against our own numbers. Every hydrogen price path is an exogenous scenario input, not a function of the deployment the model itself produces. A world in which hydrogen wins the dispatch across a large part of Europe is a world with a hydrogen economy large enough to supply it, and that scale would move the delivered price by learning and by competition for the same electrolytic output from industry and transport. We do not couple the two. The omission is defensible on materiality, since hydrogen reaches at most about 8.5 per cent of useful heat and 0.86 per cent of electricity in any scenario we run, but it means our price paths are assumptions about the wider hydrogen economy and not results of it.

The same omission runs on the load side, and there it is asymmetric. Non-heat electricity demand is grown by a single smooth factor standing for road transport and industry, and it carries no explicit electrolyser load. Heat-pump electricity is added on top and is what drives the firm-capacity requirement; the electricity that would make the hydrogen is not. The 889 GW peak, the 278 GW firm fleet, the 4,513 TWh denominator and the 0.86 per cent energy share are therefore all computed on a 2050 power system containing no hydrogen production. Adding that load would raise the denominator, and it would raise the peak only to the extent that electrolysers run in the cold snap rather than avoiding it, which a flexible operator would not. The direction on the energy share is downward and on the firm requirement ambiguous, so we flag it as an unmodelled coupling rather than a bounded one.

The inflexible winter peak. The fourth bears directly on the headline power-sector result. The firm-capacity requirement and the missing-money verdict rest on an inflexible cold-snap winter peak. Shifting a share of the heat-pump load within a fixed pre-heating window lowers the fleet by at most about 11 per cent across the three smoothing operators we test, and the cumulative shortfall with it, so the roughly 278 GW firm fleet and the roughly €357 bn cumulative peaker shortfall are upper bounds that flexibility softens rather than overturns. The reduction stays small because only 30 per cent of the heat-pump load is shiftable at the most generous corner we test, which binds before the shift window does, and sweeping the assumed length of the cold event from two days to ten holds the reduction within 8.9 to 11.6 per cent. A co-optimised flexible-load dispatch, which would size thermal storage against prices and impose no shift window, remains future work.

The energy-only market. The fifth attaches to the 9 per cent capital recovery, which the abstract, the summary of findings and the policy implications all quote. That figure is conditional on an energy-only market in which the shortage hours are not paid at the value of lost load. Paying the three shortage hours at the model's own €8,000/MWh value of lost load, which the dispatch already imposes as a penalty and pays to no generator, lifts capacity-weighted recovery from 8.7 to about 51 per cent. Even then no market recovers in full, so the direction of the missing-money result survives; its magnitude does not, and it moves further on this conditional than on any of the four above.

Annual-average emissions. The sixth runs against our own result. The cost layer prices the heat pump's winter draw at an hourly scarcity price, and the emissions layer applies an *annual-average* grid carbon intensity to all heating electricity. Heat-pump peak electricity is both costlier and dirtier than the annual mean, while hydrogen's marginal emissions scale with the peaker, so the emissions accounting carries a systematic bias in favour of heat pumps. We do not quantify it, because the model holds no hourly emissions ledger. The heat-pump emissions advantage should therefore be read as an upper bound, and no emissions-based argument here should settle a country comparison that is close to a tie. An hourly emissions ledger is the highest-priority extension of this work after the levy-equalised comparison.

Possible network double-counting. The seventh runs the other way. Retail electricity prices already embed today's network tariff, so adding an incremental reinforcement charge on top of that tariff may double-count part of the network cost of electric heating, and would in that case overcharge the heat pump against a hydrogen boiler whose delivered price carries no comparable embedded network element. The size of the overlap depends on how much of each national network charge recovers sunk assets, which an incremental charge should not duplicate, and how much is a forward-looking reinforcement signal, which it would. We do not decompose national tariffs on that basis and cannot bound the overlap, but its direction is clear. It acts against electric heating, and it points the same way as the fiscal asymmetry above, so the two qualifications on hydrogen's modelled base-load advantage compound.

What the variance screen does and does not establish. A variance-based screen over the sampled parameter space shows the 2050 emitting share to be a deterministic function of the fossil-phaseout lever, with every economic-price and technology-share axis negligible beside it. That is a property of the model's construction and not evidence of robustness, because the carbon price and the grid-decarbonisation speed are per-scenario trajectories the screen does not sample, and it does not transfer to Net Zero. The base-year demand anchor and the hydrogen boiler's non-fuel costs carry wider uncertainty than our heuristic ranges convey.

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